Don't duck around: Clarifying the roles of induction, deduction, and abduction in empirical psychology.

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Abstract

Psychology's scientific ambition is inherently manifold. It seeks to formulate general laws of psychological functioning, to identify regularities that characterize particular subpopulations, and to describe aggregate patterns in populations to inform conclusions about individual persons. Yet these different inferential targets are frequently treated as though they were epistemologically equivalent. We argue that different types of claims require different types of inferential processes if we ultimately aim for a more robust and sound psychological science. We further argue that a more nuanced understanding of the continuous interplay of abductive, inductive, and deductive inference is needed. In this vein, we demonstrate that induction and deduction are better understood as a continuum and we further illustrate how inductive processes can be leveraged to foster sound knowledge generation.

Keywords: induction, deduction, abduction, inference, psychology, methodology

Word count: 17240

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IntroDUCKtion

In this conceptual article, we argue that the hypothetico-deductive framework that has been dominating psychological science for the past decades needs reconsideration. We claim that most psychological science would benefit from a systematic debate about what forms of inference are already being applied (empirical argument), can be applied (methodological and logical arguments), and should be applied (normative argument).

Without the proposed reconsideration, inference and generalizations in psychological science will remain untrustworthy -the opposite of what Popper hoped to achieve by proposing deduction as the main form of inference in social science. Also, in that case, the usefulness of psychological science for any personalized assessment and intervention will remain limited -in an era in which personalization is a defining driver of innovation in science, economy, technology, and politics.

We argue that in the practice of psychological empirical research, the exclusive priorization of the dominating hypothetico-deductive model over the more shunned inductive forms of inference is unrealistic, methodologically outdated, undesirable, and illogical. We base this argumentation on four rationales (see Figure 1): - the empirical argument (what inference has empirically been used), showing that many studies, research programs and researchers already use intertwined combinations of deduction, induction, and abduction, whether they are aware of it or not, - the methodological argument (how inference could be used), showing how modern statistical methods and mixed methods approaches offer novel and insightful ways to combine deduction, induction, and abduction, while mitigating their respective risks in innovative ways, - the normative argument (how inference should be done in order to achieve relevant outcomes), showing how deduction, induction, and abduction should be used intertwined in order to maximize insight, improve personalized inference and interventions, and increase generalizability,

and - the logical arguments (is it logically possible to separate deductive and inductive research cycles and does that increase logical certainty of scientific inference), showing that limiting research to nomothetic-deductive approaches leads to circular reasoning and systematically overlooked boundary conditions over time, therefore systematically undermining the certainty of the inference.

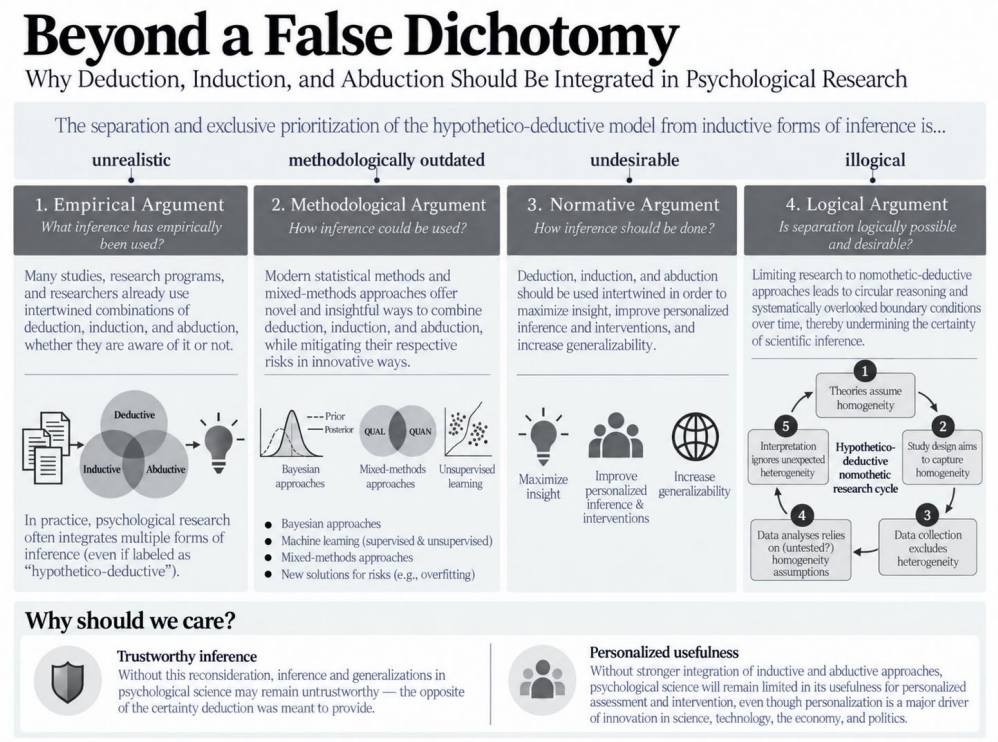


Figure 1

Why deduction, induction, and abduction are, can, and should be integrated in psychological research.

These four arguments are explained more in detail below.

The Empirical Argument: Psychologists are already integrating de- and induction, whether they know it or not

An empirical argument (what inference has empirically been used) for the need of an epistemological debate about the role of induction, deduction, and abduction in psychological science is that many studies, research programs and researchers already and often insightfully use intertwined combinations of deduction, induction, and abduction, whether they are aware of it or

not.

It has been often noted in the context of the replicability debate that many researchers merely pretend to adhere to the hypothetico-deductive model while intransparently relying on induction and/or abduction (which we may call deduction-washing) (e.g., Kerr, 1998a). So far, the proposed solution mainly focused on making it possible, desirable and mandatory to more consequently stick to the hypothetico-deductive model (e.g., Munafò, 2017).

While we support the call for more transparency and logical coherence with regard to the applied inferential processes, we argue that forcing researchers to stick to the hypothetico-deductive model is short-sided, among other reasons because hardly any psychological study is -or can be- purely deductive, and because inductive and abductive inferential processes have empirically contributed many groundbreaking, impactful, and applicable insights to psychological science. Particularly in the age of artificial intelligence, it seems out of touch with reality to pretend that no applicable or trustworthy insight can come from an inferential process that starts with lots of data, searches for patterns in these data, and then derives hypotheses about generalizable conclusions from these patterns. To illustrate this point, we provide empirical examples of research studies and programs that have maximized their insights by combining deductive and inductive elements.

The Methodological Argument: State-of-the-art methods enable and require us to integrate de- and induction

A methodological argument (how inference can be used) for the need of an epistemological debate about the role of induction, deduction, and abduction in psychological science is that many modern state-of-the-art statistical methods at least partially rely on inductive elements and were not yet available when arguments for and against the hypothetico-deductive model were widely discussed in past debates, leading to a mismatch between the methods we deem cutting-edge and insightful, and the epistemological framework that psychological science adopted decades ago. Examples for such modern statistical methods with inductive elements are

Bayesian approaches that use prior empirical data to inform their priors and/or update hypotheses or parameter estimates in light of empirical data, as well as all forms of unsupervised learning (e.g., cluster analyses and latent profile analyses, most network models in psychological science), and many forms of supervised learning (e.g., artificial neural networks).

Mixed-methods approaches are increasingly overcoming the old schism between qualitative versis quantitatie research and how machine learning and artificial intelligence help building new bridges between formerly qualitative (i.e., mainly inductive) and formerly quantitative (i.e., mainly deductive) approaches (e.g., Moeller & Schmidt, 2023).

Thus, many state-of-the-art statistical methods enable and in part require researchers to combine deductive and inductive elements. Moreover, the method development of recent decades also provides new solutions for the weaknesses of inductive inference, such as overfitting (i.e., sample dependency), thus rendering arguments outdated that in the past spoke against using inductive inference.

The Normative Argument: We gain insights, person-level validity and generalizability by integrating de- and induction

Our normative arguments (how inference should be used in order to achieve desirable results) claim that important goals of psychological science can be better achieved by reconsidering the roles of induction, deduction, and abduction in psychological science, and that, consequently, in order to achieve these goals, this reconsideration should happen.

The goals benefitting from a deliberate integration of deductive, inductive, and abductive inference processes are: - Goal 1: maximized insight (e.g., making the most of unexpected findings and exploratory insights, rather than shunning, ignoring, preventing, or deduction-washing them), - Goal 2: improved person-level explanations, predictions, and treatments, and - Goal 3: improved generalizability -and knowledge of the boundaries thereof.

Goal 1 (maximized insight) requires a reconsideration of the role of deduction, induction,

and abduction, because currently, the hypothetico-deductive model is often treated as if it was the only trustworthy, valid, or publishable path to insight. We argue that this leads to unexpected findings being underused and underappreciated in psychological science. If they are not shunned and underappreciated, exploratory and unexpected findings can be insight- and useful, sometimes groundbreaking or live-saving (for a summary, see e.g., Moeller et al., 2022). Moreover, our economies and societies are facing a major disruption due to the often unexpected, exploratory, and inductive, but often highly useful and scaleable insights provided by artificial intelligence. We argue that this development challenges psychologists' notion of the hypothetico-deductive model as the only efficient, the most trustworthy, or the most useful, instrument to gain scientific insights. In this manuscript, we demonstrate that valuable insights are missed if induction and abduction are shunned, and that consequently a wise and deliberate integration of de-, in-, and abduction is needed if scientific insight shall be maximized.

Goal 2 (improved personalization) requires a reconsideration of the role of deduction, induction, and abduction, because the theoretical and methodological approaches currently being used to achieve valid person-level explanations, predictions, and interventions all use data about individuals to inductively inform theories, hypotheses, and statistical models. With the troves of data that individuals nowadays produce, and the computing power and independence of machine learning and artificial intelligence, it is possible to develop data-driven (i.e., inductive) prediction models, hypotheses, and theories, about many individuals and their time- and context-specific behavior at large speed and scale (e.g., Yarkoni & Westfall, 2017a). In contrast, if a human researcher had to develop such theories for each individual and their time- and context-specific behavior in a theory-driven (i.e., deductive) way, it would be unfeasible in terms of speed, scale, and we cannot even start to imagine how such a theory-driven rather than data-driven theorizing about individual features would proceed. Thus, induction can contribute essential insights helping to improve personalized description, prediction, explanation, and intervention, whereas a purely hypothetic-deductive approach is of more limited use in this regard. It can be argued that we live in an era of personalization, and psychology is arguably the discipline most aiming for the

appropriate understanding, prediction, and treatment of individual persons (Moeller, 2022). Thus, there are important reasons to improve the contributions that psychological science can offer to personalization, and induction is a crucial instrument to achieve this goal.

Goal 3 (improve generalization) requires a reconsideration of the role of deduction, induction, and abduction, for much the same reasons as goal 2 (improved personalization), because both goals and their solutions go hand in hand: While psychological science has long sought to proclaim and test nomothetic laws of behavior, it has been shown logically, mathematically, and empirically that nomothetic group-level trends often do not apply to the individuals in the same samples and populations, due to heterogeneity and lacking ergodicity (e.g., Molenaar, 2004; Bryan, Tipton, & Yeager, 2021). This implies that many conclusions about individuals, drawn from general theories and their tests with group-level statistics, can be wrong for some, many, or all individuals. In other words, the generalizability of nomothetic theories and research methods to the level of individuals is often untested, and in examples in which it was tested, was often empirically refuted (e.g., Bryan et al., 2021). Many modern approaches aiming to improve both personalization and generalization in psychological science therefore tend to start out with idiographic data-driven models, then empirically identify repeating patterns across these individual models, and use these data-driven (i.e., inductive) regularities to derive knowledge about generalizability and the boundary conditions limiting such generalizability (e.g., Kuper et al., 2025), an approach called bottom-up generalization (e.g., B. Haig, 2013). The results of such bottom-up generalization tend to be more trustworthy than those resulting from top-down generalization[1], because it is at least empirically known to how many and which individuals a general trend applies or does not apply, and this distinction can also provide a useful stepping stone for the inductive search for boundary conditions. Induction is thus a crucial driver of generalization approaches.

[1] define the term top-down generalization here in a footnote

The logical arguments

Our logical arguments state that research cannot logically be purely deductive (logical argument 1), that a predominantly nomothetic-deductive research culture leads to circular reasoning and a systematical neglect of boundary conditions (logical argument 2), and that a mainly deductive approach does not guarantee the logical certainty that researchers (following Popper's reasoning) believe it does (logical argument 3).

Logical argument 1: If we consider that deduction requires a theory to deduce from, and we acknowledge that most psychological theories are build based on the prior knowledge of empirical phenomena, then that raises the logical question of whether research can ever be considered purely deductive, given that it starts with at least a partial result of inductive reasoning.

Logical argument 2: limiting research to nomothetic-deductive approaches leads to circular reasoning and systematically overlooked boundary conditions over time.

Logical argument 3: Our third logical argument states that the deductive approach does not guarantee the logical certainty that Popper hoped to achieve when proposing it.

Outline of the Manuscript

Based on these empirical, methodological, and normative arguments, we therefore argue that psychological science can, should, and already does benefit from a deliberate integration of deductive, inductive, and abductive inferential processes, and that a more nuanced understanding of their possible interplay is needed. We show that most psychological research methods, research studies, and research programs use both induction and deduction and are therefore better described on an induction-deduction continuum, rather than being labeled either purely deductive, or purely inductive. We further illustrate how inductive processes can be leveraged to foster sound knowledge generation. We argue that epistemological problems lie not merely in using induction, as students of psychology have been taught for decades, but in doing so

unwittingly and/or intransparently without deliberately mitigating known pitfalls. We furthermore argue that all forms of inference have their strengths and weaknesses, and that mitigating their weaknesses and making the most of their strengths requires researchers to be more aware of and more transparent about the exact elements of induction and deduction that they use in the various instances of a research process.

This paper is structured as follows: We give an account of the failure modes[2] of deduction and induction in the context of psychological research, concluding that it is neither possible nor desirable to conduct research exclusively relying on one or the other. This analysis is not about which inference mode is more faulty than the other, but about which inference serves what purpose under what circumstances; each inferential mode has conditions under which it is an insightful and trustworthy tool, and conditions under which it is not. We then show how describing induction and deduction in psychological research as a continuum changes how we can describe the relation of exploratory and confirmatory research, and how this can help mamking the most of unexpected and exploratory findings while mitigating known risks, such as overfitting.

We argue that this continuum view both improves the understanding, prediction, and treatment of individuals (improved personalizaton), as well as improving the trustworthiness of psychological research aiming for generalizable statements (improved generalizability).

Subsequently, we map out the limits of our proposal, and conclude with a discussion on why abduction is called for when deduction and induction have exhausted the data.

At last, we propose deliberate strategies of how to integrate induction, deduction, and, if needed, abduction, in ways that help mitigating their respective weaknesses.

Working definitions of in-, de-, and abduction and scope of this article

Deduction inference from hypothesis to data

[2] define the term failure mode here in a footnote

Induction inference from data to hypotheses

Abduction When nothing under consideration fits the data, new hypotheses enter consideration

This article focuses on the questions of whether and if, how induction, deduction, and abduction are already, can be, and should be combined in empirical psychological research. We do not question that their philosophical definitions are distinct logically; we do not aim to redefine the terms, nor offer a theoretical philosophical treaty, but instead focus on the role of induction, deduction, and abduction in empirical psychological research practices.

Our use of these terms (see Tab. 1) therefore diverges from some definitions found in the philosophical literature. Even the philosophical literature offers many, and partially conflicting and incompatible definitions, coining terms such as Peirce's "abductory induction" (Peirce, 1931; 1955, p. 152), transduction, eduction, retroduction, conduction, eliminative induction, and analogical inference. Where we diverge from definitions of in-, de-, and abduction found in the philosophical literature, we do so deliberately in order to achieve terminological clarity as well as practical applicability to empirical research.

Our use of the term deduction follows the hypothetico-deductive tradition (Hempel & Oppenheim, 1948; Popper, 1959).

Our use of the term induction departs from the Humean/Millian sense of generalization from instances (Hume, 2000; Mill, 1973) and sits closer to inference to the best explanation among an already-considered set (Harman, 1965; cf. Carnap, 1950).

Our use of the term abduction narrows Peirce's account, in which every hypothesis—not only those introduced after existing ones fail to fit—originates abductively (Peirce, 1931), and brushes against the Popperian and Reichenbachian view that hypothesis-generation admits no logic at all (Popper, 1959; Reichenbach, 1938). gladly to achieve conceptual clarity tailored to the needs of empirical psychology and rigorous definitions compatible with statistics.

Importantly, our focus on data-based inferences does not preclude nor diminish the role of inferences beyond the data of a given study. We will touch upon this issue again in the discussion.

When does deduction fail?

Deduction works best under certain circumstances

This section summarizes the failure modes of deduction, that is, the circumstances under which deduction fails to perform as it should. Do not get us wrong. We love deduction. We do it all the time. This entire article is based on arguments such as "if you want X, then you must do Y" or "if you define X like this, then Y follows". That is deduction. But this article emphasizes that psychological science ought to understand boundary conditions better, and boundary conditions do not only limit the veracity and usefulness of theories, or the prevalence and nature of psychological phenomena, but also the optimal functioning of deduction as a trustworthy form of inference.

The below addressed boundary conditions under which deduction works best are, in a nutshell: - good theories, - good measurements, - good models and concepts of causal mechanisms, and - thoughtful and transparent reflections of its weaknesses and boundary conditions and deliberate attempts to mitigate them.

Figure 2.

Deduction from theory is futile in the absence of good theory

Deduction requires and starts with a theory to deduce from. With a good theory to be tested as the foundation of deductive research, it may seem a little strange how the empirical psychological research culture following the hypothetico-deductive model has neglected the question of where good theories come from and how they can and should be constructed.

Recent debate about a "theory crisis" has revealed many reasons why psychological theories are not as good, and not as useful for deductive inference, as they could and should be (e.g., Markus I. Eronen & Bringmann, 2021).



Figure 2

Under which circumstances does deduction fail?

A central challenge for deduction are ill-defined phenomena and problems. Among the many reasons for the prevalence of ill-definedness in psychological theories are: (1) lacking clarity in definitions, lacking consensus building among scholars to agree on the best definitions, lacking clarity in theoretical work, and lacking education and incentives enabling psychologists to build logically coherent, empirically well embedded theories, as well as (2) the complex, dynamic, and time-, person-, and context-sensitive character of psychological phenomena themselves.

vague constructs

Psychological phenomena are often ill-defined, mostly due to lacking consensus among researchers and lacking cumulative science practices of researchers building on each other's work, rather than ignoring previous work when publishing own definitions and measures: Oftentimes, different researchers use the same term to refer to distinct phenomena, or different terms to refer to the same phenomenon, or both, which is known as jingle-jangle-jungle (e.g., Wulff & Mata, 2026). In addition, many psychological constructs are often multifacetted, but there is a lot of lacking consensus as to what facets exactly are included and excluded when different researchers use the same term to refer to the same multifacetted higher order construct. Moreover, researchers may believe to refer to the same construct, but then employ different items, or different measurement models, to operationalize that construct. Furthermore, there is often a gap between the often vague narrative theories and empirically testable, formalized statements, meaning that both measurement models and structure models can differ a lot when different researchers believe that they are talking about the same thing.

ill-defined research problems

A problem is considered ill-defined when its initial state, permissible operations, or goal state are underspecified (Colman, 2015; Simon, 1973). In contrast to well-defined problems (e.g., solving a formal proof, puzzles such as the Tower of Hanoi), psychological research questions frequently leave open what exactly counts as relevant boundary conditions, which variables are constitutive versus peripheral, and what would qualify as an adequate explanation (Markus I. Eronen & Bringmann, 2021).

For example, consider the research questions of what the effect of personalized learning systems on optimal learning is at school, or what the influence of large language models on students' mental health is. Solving these questions requires researchers to define the outcomes, but the here mentioned outcomes, as well as many other outcomes in psychological research, represent socially desirable phenomena, and that implies that cultural contemporary connotations, values, and political realities have an influence on how these outcomes or research desiderates should and can be defined. Thus, defining research desiderates in psychology is no completely rational, value-free process, but instead includes the art of capturing in scientific definitions what society currently wants and needs. That is neither simple, nor anything different thinkers are easy to agree on. Defining research outcomes in psychological science is inherently fuzzy. To complicate matters even more, not only the definitions of research outcomes tend to be fuzzy, but those of the predictors are as well. The personalized learning systems and the large language models mentioned in our examples rapidly develop and may today differ remendously from what they were yesterday. And even yesterday, they were more complex, and more numerous, and among them more heterogeneous, than a simple psychological journal publication would have grasped in its half-a-page definition. This implies that our definitions of predictors (here: personalized learning systems and LLMs) will often miss many aspects of the real-life phenomena, in addition to the already mentioned problem that different psychological researchers often use different definitions, are often not aware of that fact, and often fail to build on each other, leading to more jingle-jangle-jungle.

Ill-definedness poses specific challenges for deductive reasoning. Deduction requires clearly specified premises and well-delimited inference rules. When the problem space is underspecified, however, the range of admissible inferences expands dramatically, and conclusions may become underdetermined. In such cases, we often lack clarity about (a) what counts as a solution or successful explanation, (b) which boundary conditions are fixed vs. open, and (c) which representations are legitimate [e.g., the appropriate operationalization, level of analysis, or formal model; Simon (1973); Markus I. Eronen and Bringmann (2021)]. These

ambiguities are further amplified by the comparatively low degree of formalization in many areas of psychology (Meehl, 1978), which increases reliance on implicit background assumptions and untested, often not even explicitly stated, auxiliary hypotheses (Markus I. Eronen & Romeijn, 2020; Guest, 2021; Meehl, 1978).

Deduction is futile in the absence of good methods

Measurement Schmeasherment

A related and enduring issue concerns psychological measurement. Psychometrics has produced practically useful instruments for assessing individual differences and group-level patterns. However, the representational relation between the latent constructs and measurement models relying on between-person covariance on one hand, and the actual psychological processes occurring within real persons on the other hand, remains conceptually and empirically challenging (e.g., **Moeller2021?**). Insight into psychological doings is often context-bound and partially dependent on self-report, thereby introducing assumptions about introspective access, linguistic interpretation, and response processes (e.g. Tourangeau, Rips, & Rasinski, 2000). These assumptions are not always directly testable and may limit the traceability between theoretical constructs and observed data. The "measurement problem" in psychology thus concerns not only reliability and validity in a technical sense, but also the ontological and epistemological status of the constructs being measured (Borsboom, 2005).

The aim of this section is not to portray psychological science as fundamentally flawed or incapable of progress. Rather, it is to highlight the structural peculiarities of psychological phenomena and the inferential practices through which we attempt to understand them. Recognizing that our research problems are often ill-defined, that abductive and inductive reasoning are not optional extras but necessary parts of the process, and that measurement and model revision are more complex than commonly acknowledged, provides a more realistic account of how knowledge in psychology is generated. On this basis, we argue that rejecting oversimplified epistemological narratives in favor of a more nuanced view—one that differentiates

between inference modes rather than treating them as interchangeable—can improve both the transparency and the robustness of psychological research (see also Borsboom, Van Der Maas, Dalege, Kievit, & Haig, 2021).

Generally deduction can be defined as drawing conclusions from general hypothesis to specific data. Critical rationalism stipulates that if a general hypothesis does not get falsified by repeated test on data highly capable of falsifying it, the hypothesis is corroborated. This general framework is so widely accepted that some researchers equate it with the scientific method more generally (Scheel, Tiokhin, Isager, & Lakens, 2021a). Although this framework has been widely accepted, (supposedly) applied, and taught in psychology for nearly a century, the replication crisis exposed that it used to be way too easy to generate hypotheses that are accepted as corroborated but cannot withstand replication attempts.

Ill-defined concepts of causality, models and assumptions

One could hope that once researchers find a way to use clearer, more agreed-upon, facet-specifying and reality-grasping definitions of their predictors and outcomes, deduction would work just fine. However: Many psychological phenomena are inherently time-, person-, and context-specific Moeller et al. (2025), meaning a good theory would have to specify how any phenomenon or process to be explained works under any of these specific circumstances (i.e., boundary conditions) and in particular, how the many possible combinations of these boundary conditions affect that construct or process.

Furthermore, the idea that simply a better definition of predictors and outcomes will fix theory for better deduction forgets that many psychological processes are thought to be the results of complex systems, iterative cycles in which an outcome of one iteration is the predictor in the next (e.g., Vallacher & Nowak, 1994). The intricacies of such dynamical systems are little understood and while dynamical systems thinking found its way into psychological theory, testing it was long difficult until recently more intensive longitudinal data and until the tools for data analyses compatible with dynamical systems thinking started to become more available in

recent years (e.g., Failenschmid, Vogelsmeier, Mulder, & Jongerling, 2026).

As of now, much of the psychological research relies on linear thinking and linear models, assumptions of homogeneity and nomothetic trends, and and theories and practices of measurement that assume stationarity and stability in the examined constructs and processes. In contrast, dynamical systems thinking assumes that many psychological phenomena may be non-linear, time-varying and non-stationary, and heterogeneous across various boundary conditions (individuals, time points, contexts, etc.).

Deduction alone is tautological and not as trustworthy as you think it is

Deduction within the nomothetic framework is akin to circular reasoning, systematically leading to overlooked boundary conditions

The nomothetic and hypothetico-deductive research culture has over the years lead to circular reasoning, which we illustrate in Figure 3.

In the nomothetic tradition, nomothetic assumptions are encouraged and taught in university classes, easier to publish, and easier to get funding for. In line with the nomothetic one-size-fits-all approach, researchers state theories about one-size-fits-all effects, meaning they typically assume a phenomenon to be invariant across individuals, time, and contexts, until proven otherwise. Then they gather data that deliberately exclude many sources of variation, e.g., in controlled lab environments or by excluding individuals with "atypical" (e.g., left-handedness, short-sidedness, pre-term delivery, etc.) characteristics from experiments or analyses. Then they apply measures and statistical methods that rely on assumptions of invariance, i.e., homogeneity, across individuals and time. These homogeneity assumptions are often not tested, implying that if they are violated and heterogeneity is present in the data, researchers will often fail to discover it and fail to adopt their statistical and reasoning approach. If nevertheless the research accidentally reveals unexpected sources of variation, these are neither reported nor considered in the discussion of the results, due to them being "exploratory", meaning not part of the deductive approach. Thus, the research findings will reinforce the impression of one-size-fits-all effects,

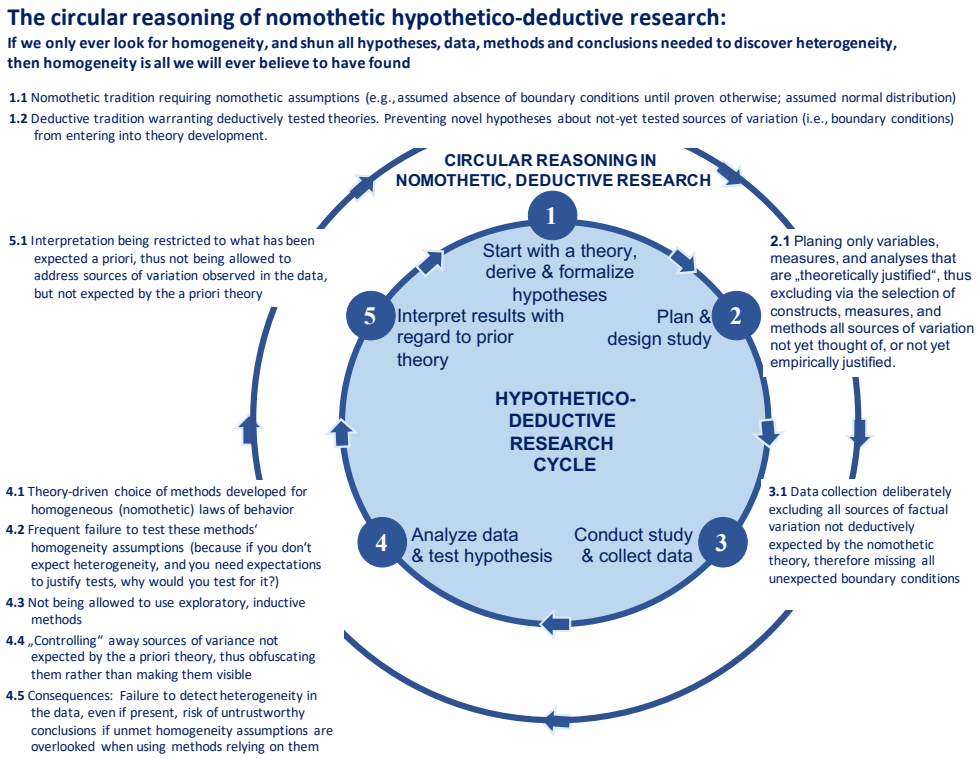


Figure 3

The nomothetic and hypothetico-deductive research culture leads to circular reasoning.

whereas sources of variation that may have been present in both the population and the collected sample are systematically understudied and ignored. With this being the end of the deductive process, evidence-based psychological research will build on such previous findings for future theory development. If a researcher dares to speculate about possible differences between groups of individuals, time points, or contexts, funders and reviewers will often require them to justify these assumptions about possible boundary conditions. But how are you going to convince your colleagues that a hypothesis of boundary conditions is worth studying, if only prior evidence counts as a good argument, if the research culture in your field has lead to a bias in one-size-fits-all results being overreported, compared to evidence about boundary conditions, and if at the same time uniform laws of behavior are considered the "sciencier" and more parsimonious assumptions until further evidence?

To give just one example of our research culture standing in the way of a better

understanding of time- and context-specific boundary conditions: One of us authors encountered a reviewer who, apparently firmly identifying with the culture of experimental empirical psychology, successfully voted to reject a conceptual article about how to discover boundary conditions, stating that there was no good reason whatsoever for experimental psychologists in a nomothetic deductive research tradition to even be interested in phenomena that are not stable over time. Noteworthy, that was after the COVID-19 pandemic had legions of psychologists rushing to study the results of the "natural experiment" of pandemic-related behavior changes, illustrating how out of this world it is to pretend that change over time was not interesting to behavioral experimental research.

Is it ever truly deductive if the theory to be deduced from was the result of induction?

Deduction implies that the a priori (before looking at data) theory was first, then the hypotheses was deduced, and then theoretical expectation was compared to empirical data to determine whether the hypotheses was a posteriori (after looking at data) empirically falsified and therefore rejected. But is there ever such a state as an a priori theory? Where do the a priori theories come from which we deduce? After more than a century of psychological science, is there ever such a thing as an a priori theory or hypothesis? Even if just consider the theorizing psychologist, pondering about psychological phenomena whilst sitting on his or her sofa, is that researcher's reasoning free of empirical knowledge about the world?

Sure, philosophers have argued about these questions for millennia, battling rationalism against empiricism, idealism against materialism. We do not intent to re-heat these theoretical philosophical debates, but let us point out that psychological science offers its own answers: Developmental psychology has shown that children develop concepts and knowledge about the world, and even language, as the result of their repeated empirical observations and experiences (Tomasello, 2009). Cognitive, social ,and experimental general psychology have shown that intuition is typically the condensation of the unconscious, as well as conscious, empirical experiences and observations that people make while living their lives, in other words: the

inductive pattern-finding in empirical data (Gigerenzer, 2023).

When constructing a new psychological theory or hypothesis, the psychological researcher employs such language, intuition, and everything he or she has learned about the world and therefore likely relies on a lot of inductive pattern finding in the process of trying to make sense of psychological phenomena. Thus, empirical data and inductive pattern finding are inherent aspects of psychological theory constructing. And that is before we consider that many researchers fill their introductory theory sections with accounts of all the results of previous empirical findings to justify their research questions, often instead of referring to actual theories, mainly stating what others found before and how that gives rise to own hypotheses. That is another form of inductive data-to-hypothesis reasoning.

Traditionally, psychologists mostly focused on theory testing, less on theory construction. Maybe that focus goes back to Popper's recommendation not to censor or otherwise limit the research questions people ask, so as to leave it up to the researcher to come up with and test every possible theory. While the liberty to discuss and test theories matters to us, we do believe that psychological science can and should focus more on the question of where good theories come from. In the era of big data, ubiquitous machine learning and artificial intelligence, we believe that informative and highly formalized theories are and will in the future often be at least in part the result of inductive pattern finding (see e.g., Lee et al., 2024).

Deduction is overrated, overstated, and overused

Deduction is futile if you only say you use it, but you don't

Deduction only works its magic if you actually do it. Just because you claim to use deduction does not mean you are. If you design research studies and statistical analysis mimicking deductive processes, but then take inductive short-cuts by doing secret p-hacking, or HARKing, then this is not deduction, and problems typically linked to induction, such as sample-specificity/overfitting, are likely to affect your data and their trustworthiness (e.g., Simmons, Nelson, & Simonsohn, 2011). Debates following the replication crisis have sufficiently

pointed that out (e.g., Munafò, 2017), we do not need to repeat it here. The problem lies not in using induction, as we will show below, but in doing so intransparently, without awareness of or attempts to mitigate its typical weakness (overfitting, i.e., unclear and likely limited generalizability). If you need inductive inference, for example because you found unexpected patterns in your data and have good reasons to believe they might be informative, relevant, and potentially trustworthy, then declare so transparently and apply or at least explicitly address in your discussion section appropriate steps to further test their trustworthiness. We explain how to integrate induction with deductive inference for such purposes further below.

Deduction is believed to fix the replicability crisis, but alone it won't

The attempts to improve psychological research methods and rigors following the replication crisis mostly aimed to tackle inconsequential applications of the hypothetico-deductive research model [see e.g., MunafoEtAl2027]. Acknowledging that researchers had long masqueraded the results of inductive pattern finding as if they were the results of deductive patterns, scientists set out to identify and prevent inductive practices such as p-hacking and HARKing. Of course, it makes a lot of sense to enable, encourage, and enforce transparency and (epistemo-)logical coherence in science. Given that most quantitative research methods were developed for and logically require the deductive approach, it makes sense to prevent researchers from claiming that they used these methods when in fact they did not. Misrepresenting the employed methods undermines readers' ability to judge their trustworthiness, and it has been demonstrated convincingly that masquerading inductive findings as if they were deductive ones has contributed substantially to the lack of replicability and trustworthiness in psychological research (e.g., Kerr, 1998a).

While we agree with the attempts to increase transparency and (epistemo-)logical coherence in psychological science, we do not believe that the full scale of the crisis to the trustworthiness of psychological science can be fixed simply by enabling, encouraging, and enforcing researchers to more consequentially stick to the hypothetico-deductive model, as for

instance (MunafoEtAl2027) propose.

The reasons are twofold: - Reason 1: Unexpected findings: The hypothetico-deductive model leaves not enough room for the value of exploratory findings, particularly illustrated by the victory march of the vastly inductive artificial intelligence and at least partially inductive pattern finding methods in data science. In psychological science, we believe that the secret wish to make the most out of unexpected and exploratory data was a driving motivation behind flawed and rightfully shunned practices such as HARKing. We do not believe that shunning exploratory research or ignoring meaningful unexpected findings fixes science. Instead, we advocate for transparent standards of how to report and appreciate the inductive or abductive insights provided by unexpected findings, combined with deliberate attempts to mitigate their known pitfalls, such as overfitting or random noise. Claiming that nothing can be learned unless a theory was first is simply outdated in the era of big data, ever-more powerful unsupervised learning algorithms and A.I. - Reason 2: Generalizability and Personalization: Recent debates about person-oriented methods, person-to-group inference, personalized assessment and intervention, and bottom-up generalization in psychological science (REF) imply that insightful, trustworthy and generalizable psychological research must often start with idiographic models and then identify generalizable trends by inductively finding patterns across many of these idiographic models (called bottom-up generalization). Due to the nowadays available ubiquitous longitudinal data, rapid innovations in idiographic and within-person statistical methods, and pattern finding algorithms, such bottom-up generalization is possible and mostly uses at least partially inductive methods. As long as the common pitfalls of inductive inference, such as overfitting, can be mitigated, such at least partially inductive methods are considered to provide more trustworthy insights about both individuals and group trends (e.g., Molenaar, 2004; Moeller, 2022). Importantly, the lacking generalizability of the traditional psychological deduction-based method toolbox directly affects the attempts to fix psychology's replicability. We argue above that due to the tautological reasoning present after decades of a deductive research culture, boundary conditions of psychological phenomena have been systematically overlooked [see also

bryan2021behavioural]. This is another way of saying: Psychological research is ripe with hidden moderators (e.g., time-, context-, and person-specificity and the many combinations thereof) that we know nothing about, i.e., have no prior empirical evidence and rather few theories about. The definition of a hidden moderator implies that it may very well be present in a sample without the researcher being aware of it. (ADD HERE AN EXAMPLE IN A FOOTNOTE) This in turn implies that the results of two studies may differ from each other, without any apparent, theorized or empirically easily identifiable reason. A researcher focusing on replicability problems may be tempted to consider this a lack of replicability, which is defined as invariance in findings despite of using similar methods and samples under the assumption of absent hidden moderators (for definitions, see Moeller et al. (2025)). Instead, our example represents a lack of generalizability, which is defined as invariance in findings because of differences in moderators (e.g., time-, context-, person-, or population-specificity). Thus, a lack of generalizability can easily be mistaken for a lack of replicability. That is made more probable by the attempts to bind researchers more strictly to the hypothetico-deductive model, because of its inherent bias to overlook boundary conditions. Thus, the solutions developed in the hope of fixing the replicability crisis may actually end up contributing to it. We cannot fix the problem with the same mindset that lead to it.

Footnote. For instance, a developmental psychologist may believe to have two samples, collected with similar methods at different times, or in different contexts (usually both), and believes that they should differ ### The argument claiming the logical superiority of deduction over induction is logically incorrect Popper's hope was that deductive inference would result in logically necessary conclusions and therefore scientific certainty. After all, the null hypothesis that all swans are white can be proven to be logically wrong, as soon as a single black swan is observed. We can trust in this conclusion, because it is logically correct. Popper wanted trustworthy science, and equated it with logically correct conclusions, as opposed to mere probability statements. By pointing out that we cannot logically prove that all swans are white, just because we have only seen white swans so far, Popper made logical veracity the gold

standard of trustworthiness and radically argued to abandon mere probability statements, such as "Probably most, if not all swans are white, based on my observation that all swans I have seen in my live - or my sample- were white".

Paradoxically, most of the social and psychological science researchers who nowadays refer to Popper when treating deductive inference as the only inference they consider trustworthy use probabilistic statements and probability-based decision rules for most of their scientific reasoning, be it in the form of frequentist significance testing, or in the form of Bayesian prior and posterior probabilities. In other words: We cling to deduction, because we believe that Popper believed that this guarantees logically correct conclusions, rather than mere probabilistic inductive conclusions, and then we use what we believe deductive inference to be all the time in combination with probabilistic conclusions, while claiming that this must be more trustworthy than induction, for the logical reason that logical veracity beats probability. Forgive us if we voice doubt that defying logic while claiming to have logic on our side is a winning strategy to increase the intended trustworthiness of psychological science, which we all aim for.

Real-Live Empirical Research Combines De-, In-, and Abduction

In Peirce's terms, deduction serves to derive necessary consequences from hypothetical premises (Peirce, 1931). Importantly, Peirce conceived deduction as operating within an "ideal state of things" (CP, exact reference needed; REF), which does not require any reference to empirical reality. However, deduction in isolation does not connect hypotheses to the world. In Peirce's mature account[3], scientific inquiry involves the iterative interplay of abduction (hypothesis generation), deduction (deriving consequences), and induction (empirical evaluation) (Peirce, 1931).

Following this view, psychological research practices, therefore, cannot be adequately

[3] define in a footnote what we consider Peirce's mature account, what would be the immature account before and what dates or what arguments exactly distinguish one from the other. Mention that Peirce's definitions evolved over time and how, and why we refer to the mature rather than the earlier works

characterized as purely hypothetico-deductive, but must be understood as relying—often implicitly—on abductive and inductive processes to bridge formal models and empirical phenomena (see e.g., Borsboom et al., 2021; B. D. Haig, 2005; Haslbeck, Ryan, Robinaugh, Waldorp, & Borsboom, 2022).

Similar suggestions of real-life psychological research often or ideally being a combination of inductive, deductive and abductive elements have been made by a number of researchers since [e.g., Theodor W. Adorno (1976a); Borsboom et al. (2021); G. E. Box (1976); box1982choice; G. Box (1999); Groot (1961); B. D. Haig (2005); Lakatos (1976); Strauss and Corbin (1994); Tukey (1980)].

Nevertheless, researchers frequently justify contemporary psychological practices by referring to a simplified or naïve hypothetico-deductive (H-D) model (e.g., Munafò, 2017). Within this narrative, theory testing is considered a linear process that derives predictions from a theory and tests them empirically. What this simplified account tends to ignore are the numerous interpretive and inferential steps that occur before and after statistical testing, as well as various inductive elements in many commonly used statistical data analyses themselves.

These include induction-informed decisions about operationalization and measurement models (REF), the (often implicit) introduction of auxiliary assumptions (Lakatos, 1976; Moeller et al., 2025), assumptions about statistical error structures and model fit (REF), and iterative processes of model revision in light of misfit or anomalous findings (e.g., Borsboom et al., 2021; G. Box, 1999; Groot, 1961; B. D. Haig, 2005). These steps are not peripheral, but essential to the inferential process. Yet, they often remain implicit when researchers claim to adopt "the H-D model". The consequence are an undercomplex account of the actual scientific reasoning, and overlooked inductive and abductive elements in that actual reasoning process.

The reasons why psychologists tend to overlook the inductive and abductive elements of inference included in their actual research practices (i.e., for the deduction-washing), can be tentatively grouped into historical, systemic, and inherent factors.

Historical Factors Why psychologists overestimate the role of deduction

Historically, psychology developed under strong influence from positivism and logical empiricism (e.g., Hardcastle, 2007) —a tradition that, despite its own substantial engagement with induction and probability (e.g., Reichenbach, 1938; Carnap, 1950), came to be read within psychology chiefly through its deductive, hypothesis-testing face. This tradition emphasized reductionism, operational definition, and the aspiration to treat psychological phenomena analogously to natural phenomena studied in physics and chemistry (Theodor W. Adorno, 1976b). The aspiration to emulate the perceived rigor, objective measurement, and formal clarity of the physical sciences [sometimes described as "physics envy"; Howell, Collisson, and King (2014); Morin (2015)] contributed to the prioritization of controlled experimentation and statistical hypothesis testing. At the same time, this orientation has been continuously criticized from the early days of the discipline onward, for example by Dilthey (Dilthey, 1990; Feest, 2007) and Wundt (see Leary, 1979), who emphasized the distinctive character of psychological and cultural phenomena, and more recently by scholars highlighting conceptual and epistemological limitations in mainstream methodology (e.g., Uher, 2021; Hanfstingl, Uher, Edelsbrunner, Dettweiler, & Gnambs, 2023). At the same time, this orientation has been continuously criticized from the early days of the discipline onward, for example by Dilthey (Dilthey, 1990; Feest, 2007) and Wundt (see Leary, 1979), who emphasized the distinctive character of psychological and cultural phenomena, and more recently by scholars highlighting conceptual and epistemological limitations in mainstream methodology (e.g., Uher, 2021; Hanfstingl et al., 2023; Moeller et al., 2025). Importantly, we take issue not with the adoption of quantitative methods per se, but with the conflation of methodological sophistication with inferential or conceptual adequacy.

Systemic Factors Why psychologists overestimate the role of deduction

Systemic factors reinforce this tendency. Academic incentive structures—publication practices, funding mechanisms, and hiring criteria—often reward empirical output and methodological novelty more strongly than theoretical integration or conceptual refinement (e.g., Smaldino & McElreath, 2016; Scheel, Tiokhin, Isager, & Lakens, 2021b). Moreover, the training of

psychologists typically emphasizes statistical methodology and research design, sometimes at the expense of sustained engagement with theory construction and philosophy of science (Borsboom et al., 2021; Markus I. Eronen & Bringmann, 2021). While methodological competence is indispensable, insufficient attention to inferential foundations may inadvertently perpetuate simplified narratives about how knowledge is generated (Oberauer & Lewandowsky, 2019).

Inherent Factors Why psychologists overestimate the role of deduction

Beyond historical and systemic considerations, there are inherent features of psychological phenomena that complicate scientific inference. Psychological processes unfold within persons, who constitute open, self-organizing systems embedded in social and cultural contexts (Bronfenbrenner, 2009; Molenaar, 2004). Such systems are characterized by high levels of complexity, nonlinearity, and context dependence (Richardson, Dale, & Marsh, 2014; Schüler et al., 2025; Van Geert, 2011). Empirical research increasingly suggests that many psychological processes are non-ergodic, meaning that population-level statistical regularities cannot automatically be generalized to intraindividual dynamics (e.g., Molenaar, 2004; Baars et al., 2025). Under such conditions, equifinality (multiple pathways leading to similar outcomes in different individuals) and multifinality (similar starting points leading to divergent outcomes in different individuals) are to be expected (Cicchetti & Rogosch, 1996). While contemporary methodologies are slowly evolving to address such complexity [e.g., intensive longitudinal designs; Bolger and Laurenceau (2013)], mainstream practice has historically relied heavily on between-person, cross-sectional designs that may not adequately capture these dynamics (e.g., Molenaar, 2004; Reitzle, 2013).

When Does Induction Fail?

ADD HERE AN INFOGRAPHIC SUMMARIZING FAILURE MODES OF INDUCTION.

Generally, induction can be defined as generating general hypotheses from specific data. Thus, induction can be used to identify patterns in observed empirical data and generating hypotheses and theories trying to make sense of (i.e., explain and predict) these data.

The main strength of inductive inference is its ability to build new theories based on what was observed in the real world. It makes sense to link the theoretical pondering about the world to some empirical knowledge of that world, instead of theorizing from an ivory tower about phenomena and effects one has never observed and experienced. Particularly in the era of big data and artificial intelligence, such inductive inference of finding regular patterns in data and using them to explain and/or predict future events has become a powerful tool, being used to optimize and predict customer experiences and consumer behavior (e.g., Ma & Sun, 2020), prevent and solve crime (e.g., Dakalbab et al., 2022), or diagnose medical diseases and psychological disorders (e.g., Abdulazeem, 2023), to name only a few groundbreaking applications of inductive inference.

Overfitting, Sample-specificity

On the other hand, induction has crucial weaknesses, and the epistemological tradition of psychological science was defined by its deliberate opposition to these weaknesses: Inductive inference is prone to overfitting, meaning an induced hypothesis will almost always fit the data on which it was trained better than any other data. Just because I have observed a regular pattern in my dataset does not guarantee that the same pattern will be observed in another dataset. Just because you have seen 100 white swans does not mean all swans are white. Furthermore, the pattern as such and the conclusion you may have jumped to based on it can be meaningless: Just because there are four letters in the first four letters of every McDonalds sign does not mean the illuminati want to convey a secret message to you. In other words: Inductive inference can be used to describe patterns in the data at hand, but its generalizability to other data, and the meaningfulness of the pattern as such, remain open questions. Until these questions have been answered through systematic cross-validation (i.e., further tests of the inductively derived hypothesis in new datasets, one could also say: replications), inductive inference is of limited trustworthiness. This problem of inductively derived patterns fitting the present data really well, but possibly not fitting other data, has been called overfitting or sample-specificity (e.g., Torralba, Isola, & Freeman, 2024). It occurs because of the inherent randomness of any data collected in the real world, meaning sample specificity due to, e.g., measurement error, hidden moderators,

sampling bias, other explainable factors, or true randomness.

Because of the overfitting problem, psychologists historically decided to avoid induction for the sake of deduction (despite of the latter coming with its own weaknesses, as we pointed out above). This historic side-taking against induction and for deduction was mainly due to the arguments published by Karl Popper (Popper, 1959), who was searching for a logically binding and therefore trustworthy form of inference. ## The historic decision to abandon induction needs reconsidering in the era of machine learning, big data, and artificial intelligence Popper's response to the overfitting problem was radical: Abandon induction altogether and reconstruct the scientific method as a purely deductive enterprise of conjecture and refutation. According to this view, scientists propose bold hypotheses and subject them to severe tests; what survives is not proven correct (because that is logically impossible), but merely not yet falsified. The appeal of this program is that appears to sidestep the overfitting problem of induction entirely, and that it emphasizes the need to test theories against available counter-evidence. However, the idea of gaining logical certainty and therefore trustworthy conclusions by abandoning induction altogether is unrealistic, misses out on many insights, and, as we have pointed out above, it is tautological and in the typical combination with frequentist or Bayesian statistics, the argument that logical certainty beats probability becomes illogical. As Hempel (1945) had already shown, even the seemingly straightforward concept of confirmation is riddled with paradoxes, and any real empirical test requires filling in quantities that the theory itself does not supply.

Pure deductivism, in practice, is a fiction: The moment a researcher estimates a variance, tests for normal distribution before choosing a statistical test, fits a curve, or selects a model over another due to its better empirical fit, induction has entered through the back door.

Lakatos (1970) tried to accommodate this by embedding falsification within research programs that are permitted a protective belt of auxiliary hypotheses, but his account still inherits the Popperian allergy to admitting that inductive reasoning does substantive epistemic work.

Meanwhile, a separate tradition—from Harman (1965) through Lipton (2004)—repackaged induction as "inference to the best explanation," conflating the ampliative generation of hypotheses with their evaluative selection and thereby muddying a distinction that Peirce had been careful to maintain. What these debates obscure is that induction is neither a monolithic operation, nor a mere embarrassment to be defined or abandoned away; it is, as G. E. P. Box (1976) emphasized, the iterative engine by which models are confronted with data, found wanting, and revised. The modern incarnation of this insight is machine learning, which has supercharged induction: algorithms that learn complex patterns from data with stunning predictive success, often without any pretense of theoretical understanding. Yet machine learning also lays bare the old pathology in its starkest form. A model with enough capacity will fit any dataset perfectly and generalize to nothing; the bias-variance tradeoff is simply the problem of overfitting restated in the language of optimization. Consider gradient descent, the workhorse of modern machine learning. Each iteration enacts this duality: the forward pass is deductive, deriving predicted values from the current parameter vector—a general statement applied to particular inputs. The backward pass is inductive, using the discrepancy between prediction and observation to revise the parameters themselves—particulars informing the general. The loss function quantifies the misfit, and the gradient attributes that misfit to each parameter in proportion to its contribution, yielding an updated general statement that will, in the next iteration, generate new deductive consequences. The cycle repeats, and with each pass the model's general claims are incrementally shaped by the particulars it has seen. It also makes visible a feature that philosophical discussions of induction tend to leave implicit: The extent of induction is not binary but graded. A model with few parameters revises little per iteration and remains close to its prior structure; a model with millions of parameters can absorb the data almost entirely, making induction so thorough that the resulting "general" statement is little more than a redescription of the particular. The degrees of freedom—quite literally, the number of parameters free to be shaped by data—determine how inductive the process is. This is the sense in which the capacity of a model is a measure of its inductiveness, and why the same formalism that corrects for overfitting in a

linear regression also, in principle, applies to a neural network: both are asking how much the data were permitted to fill gaps in the hypothesis.

What Popper attacked under the name "induction" bears little resemblance to this machinery. Enumerative induction, the inference from "all observed A's are B's" to "all A's are B's", is a limiting case so impoverished that rejecting it amounts to very little. There is no sense in which it can be partially inductive or more or less data-driven—it either generalizes or it does not. Worse, it contains no internal criterion for when generalization is warranted: a single confirmatory instance and a million confirmatory instances license the same conclusion with the same logical form. Popper was right that this kind of induction cannot bear the epistemic weight that classical empiricism placed on it. But his conclusion, that all induction is equally bankrupt, follows only if enumerative induction exhausts what induction can be. The gradient descent example above suggests otherwise. There, induction is not a single inferential leap from instances to law but a structured, iterative process with tunable parameters that determine how aggressively the data reshape the hypothesis. The question is not whether to induce but how much, and different answers to that question yield qualitatively different epistemic situations. This suggests that "induction" is not one operation but a family of operations that differ in their structure, their capacity, and their epistemic risk—and that a serious account of scientific reasoning must be able to tell them apart.

[LEVYS PART ON DIFFERENT INDUCTIONS]

Whatever form induction takes, it shares a single, unavoidable pathology: it makes us too optimistic about how well our hypotheses fit reality. Any hypothesis shaped by data will appear to account for those data better than it actually accounts for the world, because the noise peculiar to this sample has been absorbed into the hypothesis alongside whatever genuine structure it contains. The more inductive the process—the more degrees of freedom the data are permitted to fill—the wider the gap between apparent and actual fit. This is not a technical nuisance confined to statistics; it is an epistemic condition of reasoning from observation. Feynman (1974) put it

with characteristic bluntness:

> the first principle is that you must not fool yourself, and you are the easiest person to fool.

Induction is precisely the mechanism by which we fool ourselves, not through carelessness or fraud but through the very act of learning from experience. Every induced hypothesis carries, baked into its fit to the data, a component of self-deception that cannot be detected by inspecting that fit alone, with the self-deception consisting of an optimism with regard to (unclear and limited) generalizability and (unclear) meaning of the pattern as such, and that optimism is to an extent unjustified. Recognizing this is what separates a methodologically serious use of induction from a naïve one—not the avoidance of induction, which is impossible in practice, but the disciplined accounting for and deliberately mitigating of its cost. That being said, there are even examples of naïve induction worth taking up the cudgels for: Young children make sense of the world largely by making many empirical observations, detecting regular patterns in them and using these patterns to make predictions about what to expect (e.g., Köster, Kayhan, Langeloh, & Hoehl, 2020). Learning a language, including its grammar, is largely a usage-based process of being exposed to language, finding patterns and making predictions (and, here ends the inductive inference: Being corrected, noticing mistakes, and updating the pattern-based predictions with additional rules-based predictions) (see Tomasello, 2009).

When does abduction fail?

The Value of Innovation, Epiphany, and Serendipity in Scientific Reasoning

Science is about creating new knowledge. The German term for science, “Wissenschaft” reflects this main task to create (= schaffen) knowledge (= Wissen).

As every field tasked to create novel insights and solutions, science, entrepreneurship, and the creative arts, require creativity, epiphanies, serendipity, out-of-the-box thinking. What people mean by the box to think outside of is: Not clinging to the theories and expectations others had

before, and neither clinging to the obvious observable patterns in the data at hand that everybody else can see. In contrast, innovation requires approaching problems with an open, curious, independent mind and pondering about them until a novel, unheard-of solution or insight is found, often against the backlash of colleagues who find the novel idea contrary to the theories they know of (a deductive argument), or who complain that available empirical data do not (yet) let this novel idea appear to be the most probable (an inductive argument).

An important instrument of innovation is abduction as the inference to come up with novel, plausible hypotheses and theories, when new empirical observations unexpectedly contradict every available theoretical model. It relies on the art of hearing the music where others only perceive noise, for instance by recognizing the informational value of an exceptional outlier and inspecting it for the novel information that it can tell us, rather than casting it aside as error (noise) or applying ill-fitting theories to it that do not do it justice.

Groundbreaking and live-saving discoveries were made that way with abductive hypothesizing: Many if not all of us owe our lives to Fleming's unexpected discovery of the bacteria-killing features of some fungi and the subsequent development of antibiotics (see Ligon, 2004). X-Rays, post-it notes, are famously the result of the serendipity of unexpected discovery (e.g., Busch, 2023; HALFORD, 2004). As the discovery of the benzene ring is told, it included abductive thinking: The unexpected finding that benzene behaved strange for a chain-like hydrocarbon, because its six carbon and six hydrogen atoms suggested unsaturation, but it was behaving very stable. Second, the unexpected idea that a ring would explain that behavior better than the hypothesized chain-like structure, which Kekulé claims came to him when he was pondering about the problem at the fireplace and suddenly imagined the atoms forming an ouroboros -a snake biting its own tale (e.g., SELTZER, 1985). Since discovery and innovation are so crucial to the missions of science, we argue that there needs to be a valued place in the scientific epistemology and method toolbox for making the most out of unexpected findings and serendipitous discovery. Likewise, creative thinking can be taught and encouraged, or

discouraged, when training new psychologists or when evaluating new ideas in review processes for journals or funding applications.

Along with other thinkers before us (Hanson, 1979; Peirce, 1934), we propose that scientific inference, insight, and innovation require not only the analytic reasoning involved in deduction and induction, but also a generative process through which genuinely novel hypotheses, concepts, or ideas can emerge—ideas that are informed by prior theory and data, but are not logically chained to them. This generative form of reasoning, commonly associated with abduction or hypothesis generation, provides a bridge between existing knowledge and scientific discovery (Bellucci & Pietarinen, 2020; Yu, 2018). We refer to this element of coming up with innovative new hypotheses as abduction (Peirce, 1931).

For conceptual clarity, we would like to mention that our working definition of abduction follows in the tradition of Charles Peirce's distinction between abduction (coming up with novel theories explaining unexpected findings), induction[4] (seeking to evaluate existing theories by determining how probable certain empirical observations are, given a theoretical premise), and deduction (deriving expectations about expected empirical observations from theoretical premises). While both induction and deduction according to these definitions require an already existing theoretical premise, abduction is about where such new theoretical premises come from. In short, abduction is about where new theories come from, induction is about the probability of inferred conclusions, and deduction is about making logically binding, certain, inferences.

Figure 4.

[4] It is a matter of philosophical debate to what extent inference to the best explanation (IBE) should be considered inductive versus abductive, with the disagreement mostly originating in jingle and jangle fallacies of different philosophers coining different definitions of the in- and abduction and then consequently being unable to agree on their implications (REF). We firmly conceptualize IBE as induction, following Harman (1965).

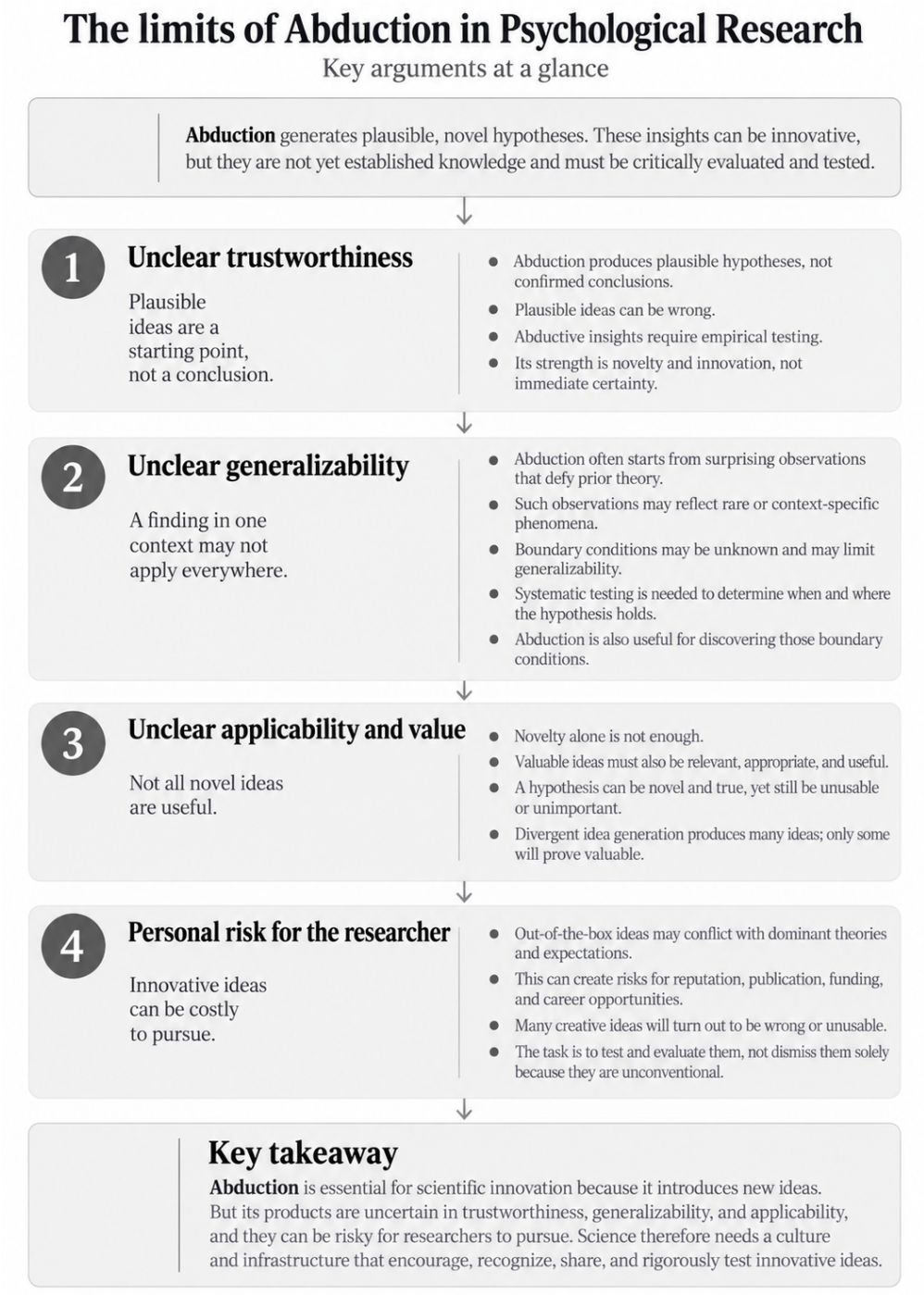


Figure 4

Under which circumstances does abduction fail?

Unclear trustworthiness of abductive inference

Abduction merely creates plausible hypotheses. Plausible hypotheses can be wrong and need to be tested. The hypotheses resulting from abductive thinking are not (yet) as such trustworthy. But they can introduce the element of novelty and innovation that progress in science, entrepreneurship and technology development need.

Unclear generalizability of abductive inference

Abduction involves the art of recognizing observations that defy prior theory and previously known empirical trends. Such unexpected observations are likely to be outliers, rarely occurring phenomena, because most other regularly occurring phenomena have already been put into theories and expectations by millennia of human observers. Therefore, the generalizability of the phenomena described in abductively derived hypotheses is not only logically unknown, it is also likely to often be limited to certain boundary conditions that were not present when others pondered about a problem and then were serendipitously present when the abductive thinker who came up with the idea pondered about it. Maybe he or she was a more creative thinker (in which case the presence and training of creative minds is a boundary condition of abductive insights), but maybe the conditions were right for the rare phenomenon to occur just when the thinker was thinking. The generalizability of abductively gained hypotheses needs to be tested systematically. As we have pointed out above, identifying boundary conditions requires and benefits from abductive thinking, so abduction both poses challenges to generalizability and is an essential tool in addressing the same challenges.

Unclear applicability and value of abductive inference

As decades of innovation and creativity research have concluded, novelty as such is not necessarily valuable, what most developers and researchers really aim for are novel and appropriate insights and solutions, meaning novel ideas that elegantly solve relevant problems in ways that communities consider valuable (Runco & Jaeger, 2012). Whether an abductively developed hypotheses adds value to scientific inquiry depends not only on whether it is testable, prevails after successful state-of-the-art deductive testing, and results in somewhat generalizable

insights. An insight can be novel and true, and yet unusable, inapplicable, or related to an irrelevant phenomenon. While that is a risk for all forms of inference, it could be argued that it tends to affect abduction more than in- or deduction, because the divergent thinking involved in coming up with new ideas tends to produce as many ideas as possible, many of which need to be thrown out before the best solution is selected from that larger pool (**eymann2026reconsidering?**).

Abduction is personally risky for the researcher

In a research culture that trains psychologists to stand on the shoulders of giants, and values article introduction sections well-embedded in prior leading theories, sticking your head out by announcing that you have had an out-of-the-box idea can be risky to your personal reputation, the article's likelihood of being published, and your funding opportunities (unless you convince one of these rare funders who value bold high-risk high-gain ideas), and thus, your career. And in fact, lots of creative minds having creative ideas will produce wrong or unusable ideas. The task is to test them and evaluate them, not to throw them out simply because they are unheard of, counterintuitive to the theories we are used to, or not yet supported by vast empirical evidence. Innovative science needs a pipeline of new ideas to be debated, tested, evaluated, and ideally an infrastructure for sharing such ideas while also getting credit for risky and demanding creative work (see e.g., Moeller & Schmidt, 2023 for such a pipeline). We would like to argue that science, due to its societal task of creating new knowledge, requires and deserves a research culture that trains, values, recognizes and supports innovation.

How can induction, deduction, and abduction work together?

Research does not stop when results come in and are interpreted, but instead is an iterative process of getting and elaborating on new ideas, developing and testing theories and hypotheses, observing expected and/or unexpected data, and pondering about the expected and the unexpected results both with existing theories in mind and with an open mind for possible unknown phenomena. Such iterations and cross-inference feedback loops have been described by many researchers, but mostly in rather general and vague models. We propose a more

fine-grained framework illustrating how induction, deduction and abduction can work together (see Figure 5).

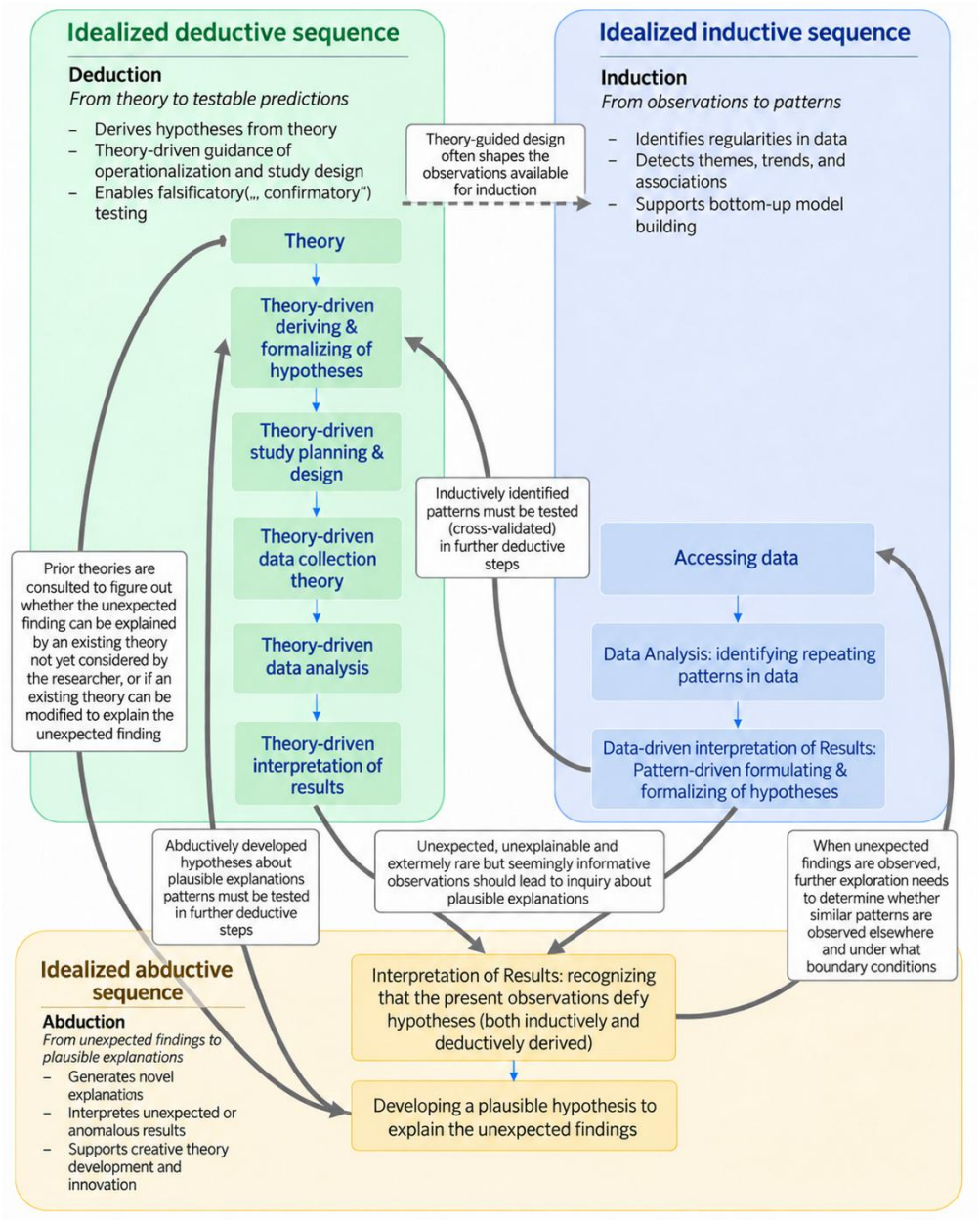


Figure 5

How can induction, deduction, and abduction work together?

The foregoing definitions establish the poles of a continuum from induction to deduction; what they leave open is the question of how a single empirical study actually navigates between them. The answer is that it does so repeatedly, and in ways that most research reports render

invisible. Every study involves moments of sharp deduction and moments of genuine induction, and the inferential quality of its conclusions depend on where they fall on the continuum and with what certainty.

Consider a study of psychological resilience. Suppose a research team, drawing on a stress-buffering model, predicts that individuals high in trait resilience will show attenuated cortisol reactivity to a standardized social stressor, and that this buffering effect will be partially mediated by reappraisal frequency. These are specific, directed predictions derived from a general theoretical framework: the data are expected, in advance, to fall within a determinate region of possibility. The inferential character of this prediction is overwhelmingly deductive. Its degree of inductiveness in the formal sense developed above is nil, because the prediction was formed without contact with the data it will be tested against. Locking this prediction in a preregistration document before data collection begins is not merely an administrative gesture; it is the operational realization of the deductive moment. Preregistration severs the feedback loop between hypothesis and evidence, preserving the logical independence of prediction from outcome that deductive inference requires. The preregistered hypothesis functions, formally, as the general premise from which a particular expected pattern is derived; its epistemic value rests entirely on that independence.

Yet even within this deductively oriented study, inductive processes are not absent—they are merely contained. The operationalization of resilience as a composite score on the Brief Resilience Scale, the selection of cortisol as the biomarker, and the choice of the Trier Social Stress Test as the stressor each reflect prior inductive sedimentation—theory-laden choices, but theory that itself bears the marks of earlier data contact. The deductive moment of the study is, therefore, always embedded in a larger inductive history, and the premises of the preregistered prediction are not purely a priori but are themselves products of prior induction. This is the normal condition of empirical science, and it does not impugn the value of preregistration. What preregistration achieves is not the elimination of induction from the confirmatory stratum but the

fixing of its scope: the general premises are held constant for purposes of this particular test, rather than left free to absorb the current data.

But the inductive exposure of the preregistered stratum does not end with its historical antecedents. Even the most conscientiously preregistered prediction often commits the researcher to acts of induction before a single observation is made. In our example, the direction of the predicted effect may be deductively fixed, but its magnitude will be induced. Should our example conclude that the Trier Social Test strongly influences cortisol with Cohen's D = .4, this conclusion is not entirely based on theory but the exact estimate certainly induced from data. The fact that the estimate is induced and not deductively predicted in this example means that it is somewhat vulnerable to being specific to this sample despite being preregistered. This means that overoptimism is not a risk that preregistration eliminates; it is a structural consequence that preregistration makes visible and bounded. The inductiveness of the confirmatory stratum is certain, and hence its overoptimism is certain. What preregistration provides is not immunity but calibration: the researcher and, crucially, the scientific community can read the preregistered document and reconstruct, at least approximately, the degrees of freedom that were open at the moment of commitment—the number of parameters left to be filled by the data, the tests that were committed to versus held back, the analytic decisions that were or were not foreclosed. The community's epistemic position is, however, always worse than the researcher's. The researcher knows which auxiliary decisions they made, which operationalizations were reconsidered during data collection, which covariates were included based as much on data as on theory. The reader knows only what the preregistration document says. Overoptimism in the confirmatory stratum is reasonably certain for both, but its precise magnitude is known better by the researcher than by anyone reading their report.

Our hypothetical example might also include some planned exploration. Suppose the research team specifies in advance a set of secondary questions, not predictions, that the data may be able to illuminate: Does the buffering effect of resilience differ as a function of prior adversity

exposure? Do individual differences in the temporal dynamics of cortisol recovery, rather than peak reactivity, carry additional variance in outcomes? Are there subgroups of participants for whom the stress-buffering model fails to hold? The planned exploratory stratum raises the inductiveness one level, and changes its character. Here it is declared in advance that the data will be used to inform the hypotheses but the extent is much more uncertain. Instead of committing to a specific statistical model, the research team specifies which questions will be asked and also does not commit to which answers they expect. By specifying which questions will be asked of the data before the data are seen, the researcher delimits the hypothesis space within which induction will operate and thereby prevents the most severe form of inductive overfit, the post-hoc proliferation of questions retroactively tailored to observed results (Kerr, 1998b). The planned exploration is more inductive than the preregistered analysis, but it is not unregulated induction. Its degree of inductiveness is higher: more degrees of freedom are permitted to be filled by the data. Given that the number of analyses for each research question they may entertain is infinite, the inductiveness is only bounded by the ingenuity and grit of the researchers, so that reporting transparency is paramount to estimate overoptimism with as little uncertainty as possible. For the scientific community reading the report, the situation is again worse. A reader can verify that a finding was flagged as planned exploration, but cannot reconstruct whether the set of planned questions was itself shaped by prior data inspection, informal piloting, or the researcher's background intuitions about what the data were likely to show. The epistemic distance between the researcher's uncertainty and the community's uncertainty is larger here than in the confirmatory stratum, because the planned exploratory commitment is more loosely specified and therefore more susceptible to undisclosed shaping.

The unplanned exploratory stratum completes the sequence. Here no prior commitment was made, and the degree of inductiveness is not merely high but formally undetermined: the researcher cannot reconstruct, with precision, how much of the hypothesis space was searched before the reported pattern was selected for presentation. Overoptimism is not merely certain, as in the confirmatory stratum, nor merely uncertain in its magnitude, as in the planned exploratory

stratum—it is unknown, in the sense that neither the researcher nor the reader possesses the information needed to bound it.

The practical implication is that researchers must maintain a running awareness of where, at any given decision point, they stand on the deductive–inductive continuum—and that this awareness cannot be reconstructed after the fact by researcher or reader.

The role of in-, de-, and abduction in personalized personality psychology

We live in an era of personalization and in this article, we argue that such personalization requires and offers new approaches to integrating deductive, inductive, and abductive inference. Personalization is a key topic of current-day economic, technological, political and scientific innovation: For example, Large Language Models personalize the support and media creation for every user up to the point that some users feel better understood and loved than even by their immediate family or therapist (Howcroft et al., 2025; Zhang, Zhao, Hancock, Kraut, & Yang, 2026). In education, personalized learning systems provide individually tailored learning tasks and materials to the right student at the right time, using A.I. to match vast pools of differentiated instruction materials and vast troves of data about students, instruction materials, past matchings between the two, and these matchings' outcomes (e.g., Nörenberg, Lazarides, Dietrich, & Moeller, 2022; Wang et al., 2024). Personalized medicine develops personalized treatments for the specific needs and physiques of individuals (e.g., Aronson & Rehm, 2015; Dugger, Platt, & Goldstein, 2018). Interactive computer games personalize the gaming experience up to the point that no two gamers' experiences and no two games are alike (e.g., Thue, 2007; de Lima, Feijó, & Furtado, 2018; Shaker, 2010). Political campaigns increasingly use personalized advertisement to try shifting their target's political attitudes, decision-making and behavior (Decker, 2025; Simchon, 2024; Tappin, Wittenberg, Hewitt, Berinsky, & Rand, 2023). If you think your muesli has too many nuts and not enough fruits, there are websites where you can choose exactly the ingredients in exactly the combination and ratio that you individually desire, and then personalize the packaging nicely with your name (https://nl.mymuesli.com/de/pages/mixer). All of these forms of personalization

have seen tremendous development in the last two decades and are driving scientific, economic, and political innovation and development. One would expect psychological science -in particular personality psychology- to be at the forefront of such personalized inference and interventions, given that psychologists tend to define their discipline as the science that explains, predicts, and intervenes on individual cognition, affect and behavior. These goals, however, conflict with the nomothetic epistemic tradition seeking general laws of behavior. Since much of the existing epistemological work referred to by psychologists was developed for and in this nomothetic tradition, we argue that some reconsideration is needed to develop a useful and logically coherent epistemological framework for the era of personalization. In recent years, psychological science has started to address aspects of personalization from various theoretical, methodological and applied perspectives (e.g., Fried et al., 2017; see also Baltes & Nesselroade, 1979). However, we argue that a systematic discussion of whether and if, how such approaches to personalization can be reconciled with the nomothetic and hypothetico-deductive epistemological framework are still missing, and a sound epistemological framework is essential for further theoretical, methodological, and applied work on personalization. More precisely, we argue that a personalized personality psychology needs (1) Personalized theories, (2) personalized assessments, (3) idiographic statistical methods in combination with bottom-up approaches to generalization, and (4) personalized interventions.

Figure 6.

Personalized theories: Openness to Individual Causal Mechanisms

In psychology, causal inference typically relies on research methods that do not necessarily generalize to the level of individuals, including group comparisons in experiments, (random intercept) cross-lagged panel models, and estimations of average treatment effects in intervention studies (e.g., Bailey et al., 2024). Personalized personality research cannot solely rely on such group-based inferences, as recent debates about person-oriented methods have shown (e.g., Moeller, 2022; Molenaar, 2004). Personalized causal inference is needed both in order to

CAUTION: This table was largely A.I.-generated. IT NEEDS EDITING & FACT-CHECKING!

Table x. The roles of induction, deduction, and abduction in psychological personalization, selected examples

	1. Personalized theories	2. Personalized assessments	3. Idiographic statistical methods & bottom-up generalization	4. Personalized interventions
Deduction	1.1 Starting with a general theory and examining whether its predicted processes hold for a particular person, including where the individual departs from the general pattern. For example, a theory may imply that a time-varying factor (e.g., sleep deprivation) changes how strongly a stressor affects mood; intensive longitudinal data can be used to test this predicted within-person relation and its person-specific variation. This corresponds to top-down, theory-constrained testing at the within-person level (Molenaar, 2004; Speyer et al., 2024).	2.1 Starting with an established construct or measurement model and testing whether its expected structure also holds for repeated measurements within persons. For example, an established depression–anxiety–stress instrument can be evaluated with a longitudinal factor model to determine whether the intended constructs are measured reliably at the within-person level rather than assuming that between-person psychometrics automatically generalize to intraindividual change (Bond & Wickham, 2023; Molenaar, 2004).	3.1 Specifying theoretically expected contemporaneous, lagged, or moderated within-person relations before estimation and then testing them with intensive longitudinal models. Dynamic structural equation models can, for example, test whether a theoretically predicted time-varying moderator changes the moment-to-moment association between two psychological states (Speyer et al., 2024).	4.1 Selecting an intervention or mechanism on the basis of prior theory/evidence and testing the predicted effect in the individual. In an N-of-1 design, one or more treatments can be prospectively compared within the same person to test which option produces the expected change for that individual, rather than extrapolating an average treatment effect directly to the person (Davidson et al., 2014; Davidson et al., 2018).
Induction	1.2 Identifying regularities in intensive longitudinal data and using them to build person-specific descriptions or theories from the bottom up. Daily or momentary data can reveal structures that differ from between-person models and can be highly idiosyncratic across individuals; recurring person-specific patterns can then be compared across persons to identify shared regularities (Wright et al., 2015; Molenaar, 2004).	2.2 Building assessment content from variables, stimuli, or contexts selected for their relevance to the particular individual, and learning empirically which indicators are most informative across repeated observations. Idiographic assessment explicitly permits variables to be individually selected or tailored, while ambulatory assessment can combine such person-specific content with common items for comparability (Haynes et al., 2009; Wright & Zimmermann, 2019).	3.2 Using data-driven person-specific time-series models to discover contemporaneous and lagged relations with relatively few prior constraints. For example, unified SEM/GIMME can recover shared paths while also estimating additional individual-specific paths, thereby identifying heterogeneous dynamic structures that would be obscured by one model imposed on everyone (Gates & Molenaar, 2012; Wright et al., 2015; Lee & Gates, 2026).	4.2 Learning which intervention option works best for an individual from repeated treatment–response data. N-of-1 trials compare treatments within a person, while micro-randomized trials repeatedly randomize intervention components to estimate proximal effects and time-varying moderation; these data can be used to construct increasingly individualized decision rules (Davidson et al., 2018; Klasnja et al., 2015).
Abduction	1.3 Starting from a surprising or insufficiently explained person-specific observation and generating a plausible new mechanism that, if true, would make the observation intelligible. The new explanation is not accepted as established; it becomes a candidate personalized theory that should generate further testable predictions. This is the core role of abduction in scientific theory construction described by Haig (2005, 2008), applied here to idiographic psychological phenomena.	2.3 Treating unexpected measurement behavior as a clue that an indicator may have an idiosyncratic meaning for the person. For example, if an item behaves differently from the other indicators of its intended construct in repeated within-person measurement, one may hypothesize that its personal meaning or context differs and then revise or supplement the individualized assessment before testing the revised measurement model. Idiographic assessment provides the personalization framework; abductive hypothesis generation provides the inferential step (Haynes et al., 2009; Wright & Zimmermann, 2019; Haig, 2005).	3.3 Treating an unexpected edge, missing expected relation, change point, or context-specific effect in a person-specific time series as a phenomenon requiring explanation. A new variable or mechanism can be proposed to explain the anomaly and then included in a subsequent measurement burst or confirmatory model. This follows the abductive sequence of detecting an empirical phenomenon, generating an explanatory hypothesis, and subsequently appraising it (Haig, 2005, 2008), while person-specific dynamic modeling supplies the empirical anomaly (Wright et al., 2015).	4.3 Using an unexpected treatment response to generate a revised person-specific mechanism. If a theoretically appropriate intervention does not produce the predicted change—or produces an idiosyncratic adverse or beneficial response—the clinician/researcher can formulate an explanation for that discrepancy and test a revised intervention prediction in subsequent within-person phases. N-of-1 methodology is explicitly designed for heterogeneous and idiosyncratic treatment responses; abduction characterizes the explanatory hypothesis-generation step (Davidson et al., 2014; Davidson et al., 2018; Haig, 2008).
Examples of possible integration of -ductions	1.4 Iterative cycle: inductively identify a person-specific regularity; abductively propose a mechanism that could explain it; deductively derive predictions; and test those predictions in further within-person observations. Failed predictions or new anomalies trigger theory revision. This integration links the idiographic requirement to study intraindividual processes directly with an abductive account of theory generation and subsequent testing (Molenaar, 2004; Haig, 2005).	2.4 Begin with theory-based common items, add individually relevant indicators, examine repeated data for informative or anomalous item behavior, formulate explanations for person-specific deviations, and then test the revised measurement model prospectively. This integrates nomothetic comparability with idiographic relevance, as advocated in ambulatory and idiographic assessment (Haynes et al., 2009; Wright & Zimmermann, 2019), while Haig (2005) supplies the abductive link between anomaly and explanatory hypothesis.	3.4 Combine common and person-specific modeling: estimate relations that replicate across many individuals, add paths needed for particular persons, interpret unexpected individual relations as prompts for new hypotheses, and test revised models on new data. GIMME operationalizes the shared-plus-person-specific statistical component by using common pathways as a backbone and adding individual pathways; abductive reasoning supplies a principled interpretation of surprising additions (Gates & Molenaar, 2012; Lee & Gates, 2026; Haig, 2005).	4.4 Use theory to define candidate intervention components and proximal outcomes; collect repeated randomized or N-of-1 treatment-response data; learn person- and context-specific effects; use unexpected successes or failures to revise the assumed mechanism; and prospectively test the revised decision rule. N-of-1 trials and micro-randomized trials provide empirical designs for this iterative personalization process (Davidson et al., 2018; Klasnja et al., 2015), and abductive reasoning characterizes the mechanism-revision step (Haig, 2008).

Note. Please note that these are merely examples of current roles of in-, de-, and abduction in personalized psychology research. The list is not exhaustive, other roles are possible. We merely mean to show that all three forms of inference contribute important aspects to the personalization workflow. This table was largely completed by ChatGPT (ChatGPT (GPT-5.6 Sol; OpenAI), accessed October 2, 2026). The authors provided the structure of the table, the prompts documented in Appendix X, and edited and fact-checked the ChatGPT-completed table. The authors would like to use this example opportunity to point out to the readers that the arguments made in this manuscript are so established (despite of being largely ignored among psychologists), that ChatGPT summarized the role of in-de- and abduction unprompted with the sentence: "The strongest personalization workflows typically cycle among all three rather than using one inference mode in isolation." It appears to us that if psychologists don't start reconsidering their very limited nomothetic-deductive epistemology, ChatGPT and other LLMs are already now able to provide a logically and empirically sounder analysis of the inferences used and needed in psychological research. And they will do that largely based on inductive inference, meanwhile many of our colleagues continue to claim that the only best way to insight is deductive. Automatically ChatGPT-created notes to this table: "Examples were revised so that each is anchored in peer-reviewed literature. The cited authors do not necessarily describe their procedures as deduction, induction, or abduction; the inference labels are a philosophical interpretation of the research operation. For abduction, Haig (2005, 2008) provides the methodological basis: surprising or insufficiently explained phenomena motivate generation of plausible explanatory hypotheses, which then require further testing."

Figure 6

The roles of in-, de-, and abduction in personalized personality psychology

understand, predict, and effectively treat individuals, and in order to build causal models bottom-up by comparing many idiographic causal models with each other to identify the causal mechanism that repeat, or do not repeat, across individuals, as well as their boundary conditions. We envision such personalized causal inference as the art of developing and testing idiographic models of causal mechanisms. Rather than limiting our definition to granger causality (DEFINE HERE IN A FOOTNOTE), we define idiographic models of causal mechanisms as micro theories that explain and help predicting which exact antecedents lead to which exact outcomes through which exact mechanisms under which circumstances. Such theories need to be plausible, logically coherent, falsifiable, tested or about to be tested, they should both fit to observed empirical data (inductive support) and withstand extensive testing against any thinkable counter-evidence (deductive test). They should be in line with what else is known about laws of nature (e.g., Physics, Biology), or if they are not, the possible reasons for that discovery should be discussed and further studied (abductive elaboration). In order to explain, predict, and effectively treat

individuals, we need theories that address individual features, even -or particularly- if they deviate from overall group trends. We need theories, data, and statistical methods that clarify within-person variability across time points, across different contexts, and other boundary conditions of behavior, cognition, and affect. We most likely need inductive data-to-theory inferences in order to build theories and prediction models to explain and predict individuals and to make the most of the increasingly available intensive longitudinal data about individual behavior, cognition and affect from e.g., wearable devices and their sensors, clickstream data from individuals' behavior with computers.

Personalized assessments: Openness to idiosyncratic or rare patterns

Understanding individuals and their heterogeneity means that personalized assessments preparing for personalized interventions need to be open to discover novel features of the functioning of an individual that do not necessarily represent the general group-level patterns observed in larger groups of other individuals. For instance, just because workplace engagement and work-related burnout symptoms tend to be negatively correlated across individuals does not mean that every individual person experiences high values in one and low values in the other variable (e.g., Moeller, Ivcevic, White, Menges, & Brackett, 2018; Salmela-Aro, Moeller, Schneider, Spicer, & Lavonen, 2016; Tuominen-Soini & Salmela-Aro, 2014). While in this example, the joint experience of high values of both engagement and burnout occurs so often that a latent profile analysis (LPA) discovers the subgroup of the roughly 25-33% of individuals showing this pattern, it is also possible that other phenomena occur so rarely, or even uniquely, that no subgroup analysis like LPA would discover it. For example, some psychological syndroms are so rare that they are very unlikely to show up even once in any random sampling with only a few patients ever being observed worldwide (e.g., Iqbal, Prasad, & Ritey, 2014; Querter et al., 2025). As a nomothetic researcher searching for generalizable laws, you may find these few individuals neglectable, but as their parent, as the affected patient yourself, as their treating psychologist or psychiatrist -nomothetic training or not-, or as a curiously inquiring researcher hoping to discover new phenomena that others have not yet addressed, you would likely think otherwise.

To give further examples of the informational value of rare or even unique occurrences, imagine you are a crime investigator. The fact that your suspect shows recurring regular behavior, such as having their morning tea each day at the same time before walking the dog may be interesting, but you may be even more interested in sudden, never before see behavior, such as the suspect withdrawing all their savings, or booking a one-way ticket for a journey despite of rarely traveling before. In this example, the unique deviation from the repeating behavioral trends is not just noise, but may even be the deciding detail to crack a case, or to prevent a terrorist attack. Likewise, if you are the treating psychotherapist of a patient who tends to experience relapses of severe psychopathological symptoms (e.g., psychosis, manic episode, eating disorders) whenever their hard-fought routine is disrupted, a unique event, such as the workplace closing down, the patient graduating from school and having to find a job, or the birth of a child, may be a reason to pay heightened attention to the patient. All of these are examples of rare, even unique features of time, context, or individuals, that carry information highly relevant to a save and effective treatment or intervention on individual persons. In order to be useful for personalized interventions, assessments need an openness to capture such features of situation, context, and individuals, that matter most to the individual in question, and that ultimately predict their behavior and the efficacy of a personalized intervention best.

Personalized statistical modeling and bottom-up generalization: From the individual, to the group trend, not the other way around

The need for within-person methods and bottom-up generalization: Personalized personality psychology needs idiographic and within-person statistical methods (see e.g., Moeller, 2022; Molenaar, 2004). Where research aims to identify generalizable trends, this generalization should be conducted bottom-up (see e.g., Moeller et al., 2025) in order to be trustworthy and because that facilitates the search for time-, context-, and person-specific boundary conditions. The need and impossibility of testing all homogeneity assumptions: In psychological science, where most studies employ statistical methods relying on (often untested) homogeneity assumptions, such as unimodal normal uni-, bi-, and multivariate distributions, it is important to

keep in mind that such homogeneity assumptions represent themselves theories about what data to expect, and as a priori theories, they need to be tested. A problem for personalized and other personality psychology alike is that the more complex, multilayered and boundary-rich data get, the more homogeneity assumptions need to be tested. The intensive, contextualized longitudinal data that are typical for personalized research are typically such complex data full of possible sources of (e.g., time-, context-, person-specific) heterogeneity. As we have demonstrated (Baars et al., 2025), it is not trivial to test all of the relevant homogeneity assumptions underlying the person-oriented methods needed for personalized research. It seems unlikely that anyone is currently testing all of the relevant homogeneity assumptions in person-to-group bottom-up studies. We even argue that testing all homogeneity assumptions is essential as it is impossible, because homogeneity needs to be demonstrated across possible sources of heterogeneity, and the hidden moderator problem mentioned above implies that researchers can never be sure of the absence of the moderators they have not considered yet, while working in a research culture that has a known bias to overlook them. Bottom-up does not only mean from individual person-specific patterns to general patterns that repeat across individuals. It also means from the unique observation in one context at one time to the discovery of certain patterns that repeat -ordo not repeat- under certain boundary conditions. Ergodicity is not just about individual person to group of persons, it is also about individual time point to group of time points or context to group of contexts.

Personalized Interventions

Personalized interventions are often developed using a combination of deductive, inductive, and abductive inference. Inductive contributions to personalized interventions: For any interventionist aiming to scale up personalized interventions to a number of individuals, it is a time- and resource-demanding challenge to develop for each person the required personalized tools: Personalized theories and personalized models of causal mechanisms, personalized assessments, personalized statistical models. For such considerations of intended scalability with limited resources, many personalized interventions will rely on personalization processes that can

be automatized and performed by computers, and these include typically inductive data mining procedures that estimate idiographic models in parallel, then compare them to bottom-up identify patterns that repeat across individuals (and/or time points, and/or contexts). An example for such personalized interventions developed with the help of inductive pattern-finding is the study by [mishra2021detecting], who used machine learning models to detect and improve each person's receptivity to the intervention in the natural environment, so that based on each person's individual previous pattern, the timing of the intervention can be optimized to maximize the intended outcomes. Deductive contributions to personalized interventions: In addition, developing personalized interventions intended to be scaled up to a number of individuals will likely make use of some aspects of deductive reasoning. There will usually be general theories detailing core components, causal mechanisms, predictors and outcomes of the psychological phenomena to be intervened on. While it will often be unclear whether they apply to the actual individual at hand, they can still often serve as checklists or guidelines what to look out for and what to test for the present individual in a deductive approach. In addition to specific psychological constructs, theoretical knowledge of general design principles determining the efficacy of personalized interventions can deductively guide the development of novel personalized interventions. Moreover, any result of inductive or abductive inference needs to be cross-validated and thus, deductively tested, before its (local or global) generalizability can be assumed and overfitting ruled out. Thus, deductive contributions to developing personalized interventions can add informational value both before, and after, inductive steps. Examples for such existing theories deductively informing and guiding the development of personalized interventions are the frameworks specifying the general components, design principles, decision rules, and other features determining the efficacy of just-in-time ambulatory interventions (JITAIs) (see e.g., Nahum-Shani et al., 2018; Nahum-Shani & Murphy, 2026). Abductive contributions to personalized interventions: Identifying the best treatment for the best person at the best time and in the suitable context requires extensive knowledge about the time-, context-, and person-specific boundary conditions -and combinations thereof- that determine the efficacy

of an intervention. As we have stated above, the search for boundary conditions needs to remain open to new discoveries (i.e., abductive inference), for one reason because of the hidden moderator problem (it being logically impossible to rule out that an as of yet unknown boundary condition may limit the efficacy of an intervention in the studied population, and for a second reason, because an individual, a time period, or a specific context, may defy the trends and patterns seen anywhere else. Remaining open to the possibility that an intervention that has worked before for other individuals, or for the same individual under different conditions, may turn out to be inefficient, is nothing else than remaining open to abductively discover and take seriously a misfit between everything expected and an unexpected novel observation not yet explained by existing theories (deductive inference), nor generalizable data patterns (inductive inference). As with every rare occurrence, the challenging question is whether it represents meaningless noise or an informative signal, meaning whether it can teach us anything trustworthy about our research question, that is, whether it reveals a phenomenon that can be repeated, and/or can be used to predict relevant outcomes (e.g., improvements of the symptoms the personalized treatment aims to improve). For example, a personalized intervention that had expectedly increased an individual's physical activity at the beginning of the intervention unexpectedly lost its effectiveness until the effect was indistinguishable from zero at the end of approximately one month (e.g., Klasnja et al., 2019). Also unexpectedly, only walking but no anti-sedentary suggestions had effects in this personalized intervention study. This unexpected decline in treatment effectiveness can generate the abductive hypothesis that intervention fatigue and the intervention content (walking versus anti-sedentary suggestions) represent previously unknown boundary conditions; subsequent observations can then test whether these conditions reliably predict intervention failure.

Person-level inference is a problem of generalization

We started by positing that several characteristics (ill-definedness, measurement, non-ergodicity etc.) make generalization difficult and that the replication crisis exposed this difficulty. Any inference rests on a generalization claim, and the inference is only as sound as that

claim. Some claims are broader than others and psychology is often interested in generalizing on different levels, often at the same time, from populations to subpopulations, down to the individual (and often under varying boundary conditions). The inference and the assessment of its soundness must reflect that. Any generalization claim hence raises the question: generalizing to what? As we have extensively argued, generalization claims also require us to consider the inference modes that were used to generate it. Often we observe that there is dependency between the generalization target (population vs. individual) and the dominant inference mode. Strongly deductively derived claims tend to target populations, whereas claims about individuals tend to involve more induction. This dependency is not inherent to the inferential machinery itself but on the availability of theoretical considerations and data. It is conceivable to have grasp of "general laws" that only govern a single individual. At the same time, if we have an individual in front of us but neither knowledge nor data about said individual, our best guess is to treat this individual as an average member of the population. Then we apply a deductive inference mode, because we apply general laws (from the population) to a particular (individual). As soon as we have data about the individual, we may induce changes to the general law to better describe this particular individual. This interaction of what we know about an individual and what we know about the population is hence governed by a mixture of deduction and induction. As we have consistently stressed, engaging in induction comes at a cost and requires careful calibration of the uncertainty to account for overoptimism. How to apply these principles to person level inferences is the topic of this section.

While we hold that some induction is necessary for any research in psychology, for person-level inference induction may even be the dominant inference mode, and a psychology of individuals that refused to induce would not be a cautious science of persons but no science of persons at all.

To see why, return to the epistemic predicament sketched above. Facing a person about whom we know nothing, deduction is the only available inference mode: we apply what holds for

people in general, or for the subpopulations this person visibly belongs to, and treat the individual as an instance of their reference class. This is a respectable inference, and often the best available one, but it is limited in principle. Deduction can only make explicit what its general premises already contain; it can apply knowledge of persons to a person, but it cannot yield new knowledge about that person. Whatever makes this individual differ from their reference class—the residual that differential psychology exists to study—cannot, by construction, be derived from those premises. This is what the inattentive date and the formulaic therapist of the previous section have in common: both competently apply knowledge about people in general, and both fail because that was never the kind of knowledge the situation called for.

Non-ergodicity is crucial here: where within-person structure departs from between-person structure, the reference-class prediction is not merely imprecise but wrong in kind (Fisher, Medaglia, & Jeronimus, 2018; Molenaar, 2004). If the individual is to be known as an individual, the knowing must proceed from the individual's own particulars to revised general expectations about them—which is, by our definitions, induction.

The idiographic tradition saw this long before the statistical machinery existed to honor it. Windelband (1998) distinguished nomothetic sciences, which seek general laws, from idiographic sciences, which seek knowledge of the particular; Stern imported the distinction into psychology, and Allport (1937) popularized it there as a demand—never fully met—for methods adequate to the single case (Phan et al., 2024). What that demand amounts to, in the vocabulary of this paper, is induction: general statements whose scope is a single person, earned from that person's particulars. The idiographic tradition's vulnerability was that it could describe such knowledge but not discipline it; idiographic claims were suspected, often fairly, of being unfalsifiable. Its second vulnerability was terminological: Allport himself later abandoned the terms (Allport, 1962), which have since accumulated so many meanings—between- versus within-person, quantitative versus qualitative, variable- versus person-centered—that they obscure rather than specify which generalization is being claimed (Phan et al., 2024). Our claim in this section is that both vulnerabilities are now addressable: the systematization of generalization claims by Phan et

al. (2024) supplies the missing precision, and the formal machinery that exposes induction's overoptimism supplies the missing discipline.

The overoptimism identified in the preceding sections should accordingly be read not as an argument against induction but as a consequence of induction that can and must be estimated. What person-level inference demands is not less induction but induction conducted with care: learning from a person's data without mistaking sample-specific noise for stable characteristics of the person. The remainder of this section is an account of what that care consists in.

At its core lies an epistemological principle far older than the statistical machinery: a claim induced from data can only be tested on data that did not inform it. Cross-validation is the family of procedures that institutionalize this principle within a single study; some portion of the data is withheld from the induction and reserved for asking, afterwards, how well the induced claim generalizes (Stone, 1974; cf. Yarkoni & Westfall, 2017b). The discrepancy between fit on the data that shaped a claim and fit on the data that did not is an empirical estimate of exactly the overoptimism we have been tracking. The mechanics vary, and most of them belong to statistics; what belongs to epistemology is a single insight: the choice of what to withhold from induction is the operationalization of the generalization claim. Whatever unit is held out—an observation, a stretch of time, a person, a study—is the unit to which the resulting estimate licenses generalization, and nothing broader. A resampling scheme is a small simulation of the inferential situation we claim to be in. Unfortunately it is all too easy to simulate the wrong one.

Which inferential situations there are to simulate is precisely what Phan et al. (2024) systematize. They cross the focal entity of a claim (a person or a population) with its type of generalization (none, entity-specific, or cross-entity), and each generalization type in the resulting matrix, we argue, demands a different unit to be withheld.

A cross-person generalization—the claim that a regularity holds for each person, not merely on average (Phan et al., 2024)—is tested by withholding entire persons: the model is

induced on all individuals but one and evaluated on the withheld individual (REF). Such a scheme turns Windelband's question into an empirical one: does knowledge of persons-in-general apply to this person? Where between-person structure does not transfer, prediction for the held-out person fails, and non-ergodicity ceases to be abstract and becomes measurable. The distribution of errors across held-out persons even separates the strict from the probabilistic reading of such a law: a law claimed for all persons must survive every fold, a law claimed for most persons must survive most (Phan et al., 2024). A person-specific generalization runs not from person to person but from a sample of occasions to the population of occasions in one person's life (Phan et al., 2024). The unit to withhold is therefore the occasion, and above all the future occasion: the model is induced on the earlier portion of the individual's series and evaluated on the later, with the boundary rolled forward to yield repeated estimates (Beck & Jackson, 2022; Bürkner, Gabry, & Vehtari, 2020; Tashman, 2000). A population-specific generalization—from a sample of persons to the population as an aggregated whole (Phan et al., 2024)—is what the common pooled design actually tests: observations from many people are randomly split, and the estimate answers the question of generalization to further exchangeable observations from that population. When observations from the same person land on both sides of the split, even that estimate is inflated, because within-person dependence carries information from the training data into the validation data (Karbalaie, Abtahi, & Häger, 2026; Roberts et al., 2017). And when a population-specific estimate is read as licensing claims about individuals, the resampling scheme has reproduced the ergodic fallacy in miniature (Molenaar, 2004; Phan et al., 2024): the folds were exchangeable observations, so the license extends to exchangeable observations and to nothing else. Choosing among these schemes is therefore not a technical preference but a declaration of which claim one is making. The remaining refinements—buffers that keep serial dependence from leaking across the boundary between training and validation data (Bergmeir & Benítez, 2012; Burman, Chow, & Nolan, 1994; Racine, 2000), conditions under which random splits regain their validity (Bergmeir, Hyndman, & Koo, 2018), the fragility of estimates from the short series typical of experience sampling (Liu & Zhou, 2024), the advantage of temporally ordered schemes when the process

itself drifts (Cerqueira, Torgo, & Mozetič, 2020)—are the statistical implementation of our philosophical commitment.

One cell of this matrix requires no such estimate. An entity-specific portrayal describes a particular person without any intention to generalize (Phan et al., 2024); because it claims nothing beyond the data, there is no overoptimism to estimate. The moment the portrayal is used to predict anything—tomorrow's mood, the response to a treatment—it has become a person-specific generalization like any other.

One scheme deserves particular attention because it formalizes the process of learning about a single person over time. Prequential evaluation scores a model by its sequence of one-step-ahead predictions: predict the next observation, observe it, let the observation revise the model, and repeat (Dawid, 1984). This is the therapist who forecasts how a client will respond to an intervention, observes the response, and updates the case formulation—with the one addition that every forecast is scored before the model is updated. Deduction and induction alternate here exactly as in gradient descent: the forward prediction is the deductive moment, the revision the inductive one, and the running score an uninflated estimate of how well the model predicts this particular person. Far from rendering induction suspect, the prequential scheme shows that continuous induction and valid evaluation are compatible: the model learns from every observation, but only after that observation has been predicted and scored.

The same separation between induction and evaluation must cover every decision the data inform. Whenever the induction procedure itself has adjustable settings—how complex a model, how strongly regularized, which variables enter—and those settings are chosen to optimize validation performance, the validation data have become part of the induction and can no longer provide an independent test of it; nested schemes restore the separation by tuning within an inner loop and evaluating on an outer one, both respecting the person structure (Karbalaie et al., 2026). The general principle matters more than any implementation: the data used for evaluation must remain independent of every step of the induction, tuning included. A

holdout that repeatedly informs modeling decisions is simply more training data, and the estimate it yields is inflated by the very overoptimism it was meant to measure—not because induction failed, but because the separation between induction and evaluation did.

Once the resampling scheme is understood as the operationalized generalization claim, the scale of the single study stops being special, and familiar institutions of scientific self-correction appear as the same principle enacted between studies. A direct replication is a held-out fold at the level of studies: the original study is the training fold, on which the hypothesis was—however it was presented—partly induced, and the replication is the fold on which it is evaluated. Read this way, the replication crisis was not evidence that psychology induces too much; it was evidence that psychology had been reporting training fit as if it were held-out fit—the evaluation was skipped, not the inference overused. Conceptual replications and multi-site projects extend the scheme along its natural axes: varying the operationalization, population, or context ensures that the held-out study differs systematically from the original—the between-study analogue of buffered resampling (Roberts et al., 2017)—so that the estimate reflects transfer rather than resampling error, and a many-labs project is leave-one-lab-out cross-validation in all but name (REF). The analogy also makes precise which replication tests which claim: a direct replication in the same population tests a population-specific generalization on a new sample, a replication in a new population tests a cross-population generalization (Phan et al., 2024), and neither validates person-level claims, because in both the unit held out is a sample—not a person, and not a person's future.

Meta-analysis, in turn, is what the aggregation of folds looks like at the between-study scale. A random-effects model treats studies as exchangeable draws from a population of possible studies; its between-study heterogeneity is the between-fold variance, an estimate of how far the claim generalizes across contexts; and the familiar leave-one-study-out sensitivity analysis is cross-validation at the level of studies in the most literal sense (REF). Two consequences follow. Publication bias becomes a leakage problem: folds pre-filtered by their outcomes were never

genuinely held out, and averaging leaked folds does not remove the bias. And—more fundamental for our argument—aggregation cannot repair a mismatched generalization target. A meta-analysis of between-person effect sizes estimates a population-level quantity with ever greater precision as studies accumulate; under non-ergodicity, no amount of that precision converts it into a claim about the structure of any individual (Fisher et al., 2018; Molenaar, 2004). The remedy is not simply more data but different data: person-level data, person-level splits, and, where possible, the aggregation of within-person estimates rather than of between-person aggregates (REF).

For study design, the upshot is that the resampling scheme should be chosen when the inference target is chosen, because the scheme dictates what data are worth collecting. A claim about new persons draws its statistical power from the number of persons, almost regardless of how densely each is measured; a claim about a person's future draws its power from the length and resolution of that person's series, almost regardless of how many other people are enrolled. The two are not interchangeable: a hundred participants measured twice cannot support person-specific generalizations, and one participant measured a thousand times supports no claim beyond that person. Prespecifying the scheme is the natural extension of preregistration into the inductive stratum: where the preregistered hypothesis fixes what is deduced, the preregistered validation scheme fixes how the induction will be evaluated—which units are withheld, how dependence is buffered, which decisions the data may inform. Phan et al. (2024) ask researchers to state their generalization intention explicitly and to check its alignment through design, analysis, and inference. A reader who knows which units were withheld knows which claim was actually tested, whatever the discussion section asserts. What legitimizes inductive person-level research is the estimate of overoptimism, not the restriction of induction. Such a study tests exactly one premise: that the induced claim about this person generalizes beyond the data that induced it.

The choice, for a science of individuals, was therefore never between induction and certainty. It is between disciplined induction and no person-level knowledge at all.

Table 1

Working definitions of deduction, induction, and abduction, and of deductiveness and inductiveness as graded properties.

Term	Working definition
Deduction	From the general to the particular. The process of deriving statements about particulars from general statements through logical operation. In deductive inference, if the general premises are true, the particular conclusions must necessarily be true—the entire logical content of the conclusion is contained within the premises.
Induction	From the particular to the general. The process of using particulars (observed data) to form or support general statements that extend beyond what has been observed. In inductive inference, the conclusion goes beyond what is logically contained in the evidence.
Deductive (as a property)	The extent to which predictions or statements about particulars are independent from and not influenced by the specific data they concern—the extent to which they stem purely from general principles or theories without reference to observed instances.
Inductive (as a property)	The extent to which a statement or prediction is influenced by and dependent upon specific observed data—where the particular instances shape and modify our general claims or expectations about unobserved cases.

Term	Working definition
Abduction	When nothing under consideration fits the data, new hypotheses enter consideration. For instance, unexpected findings may suggest previously unconsidered boundary conditions or unknown causal mechanisms. When none of the statements from the set under consideration about the particulars achieves a sufficient minimization of the loss function, the set is extended and induction/IBE is repeated. The extent of induction/inductiveness can no longer be formally ascertained because data has influenced the extension of the set in ways not captured by formal inductive logic. Inferences that go beyond empirical observations are not considered here.

Like modern Bayesianism our proposed approach is not (and cannot be) a solution to the problems introduced by induction but should be regarded as a way to manage inductive inference so that we can generate inherently imperfect knowledge in a controlled fashion.

In this vein, we must accept induction as a fallible but necessary pathway to (practical) knowledge.

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